

Thread - 2026-01-10

<https://x.com/RiikonenMarko/status/2009999376098197857>

1/21 - 2026-01-10 14:42 UTC

9/10 Jan 2026, Rovaniemi. At the coalface all night. So far the coldest night. My thermometer on the ground was at -39 C when I left the photo location, golf-course. -41 C in another location and I didn't even visit the coldest places.

Anyway, I was most of the night at the golf-course and it was constant ice fog. No time for sitting the in warmth of the car, just swapping the cold camera to a hot one every once in a while.

Got maybe around 400-500 photos. Even odd radii at one stage, but not as good as the other night. Here is single image, nadir arc glittering low down.

Some tense moments as the other memory card had lost big chunk of the night's photos. Sweaty hands while recovery software was progressing. Got all back so going out for the coming night's chase in much better mood now.

The night has already started, but I am late because of the card issue. And first thing to do before any chase is to go buy gaffer tape to address those lens reflections from the blocker.



2/21 - 2026-01-23 14:24 UTC

9/10 Jan 2026, Rovaniemi. Time to start untangling the results from this coldest night of the winter. This was in the Sierijärvi area, a short swarm outside of the main plume. Two max stacks of 4 and 23 frames and 23 frame ave stack. Nothing to comment, just basic stuff.

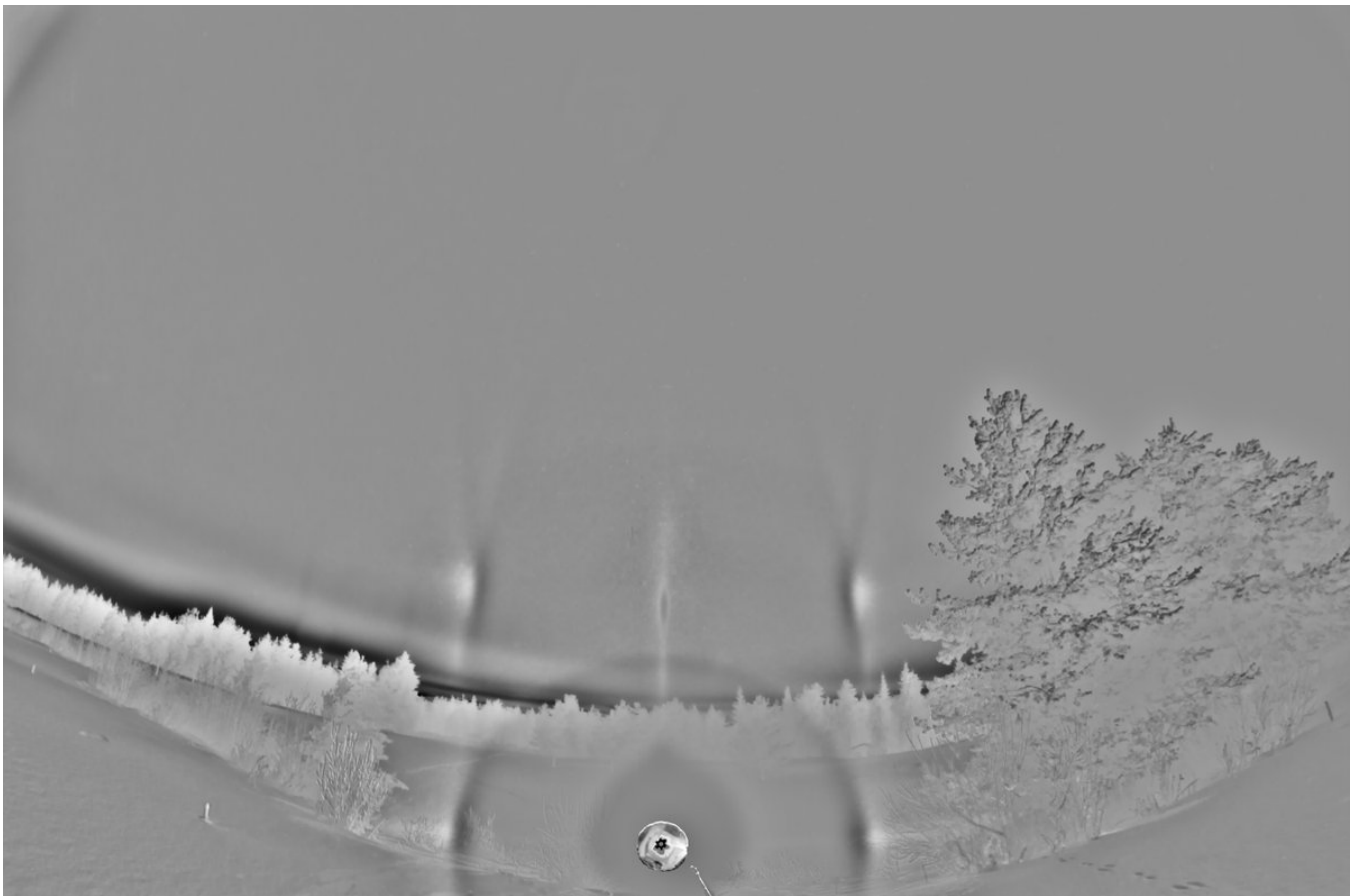


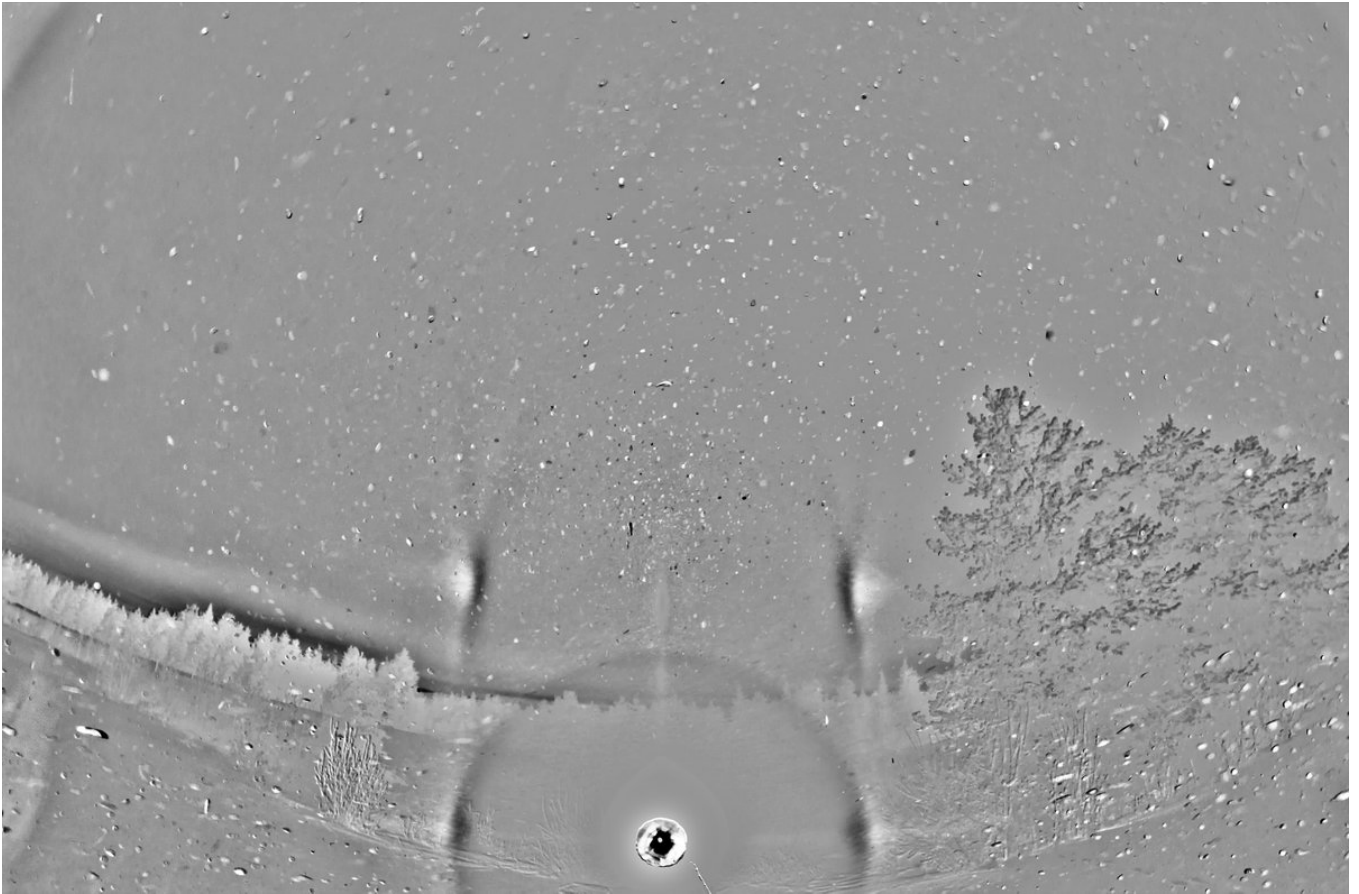


3/21 - 2026-01-23 14:37 UTC

9/10 Jan 2026, Rovaniemi. This and the rest of the images are from the golf-course, so more light pollution. What are those spots underneath the counterpartaria? Anomalous brightenings in Liljequist arcs? Ave b-g, max b-g, ave bgr, ave usm. 25 x 30s.







4/21 - 2026-01-23 14:48 UTC

9/10 Jan 2026, Rovaniemi. There was an unceasing churn of diamond dust at the golf course. At one point the counterparhelia became fuzzy and expanded inwards. I presume this was due to counter-18° plate arcs.

[video: 2014711769273454755-07ebZPmfetAMq_9l.mp4]

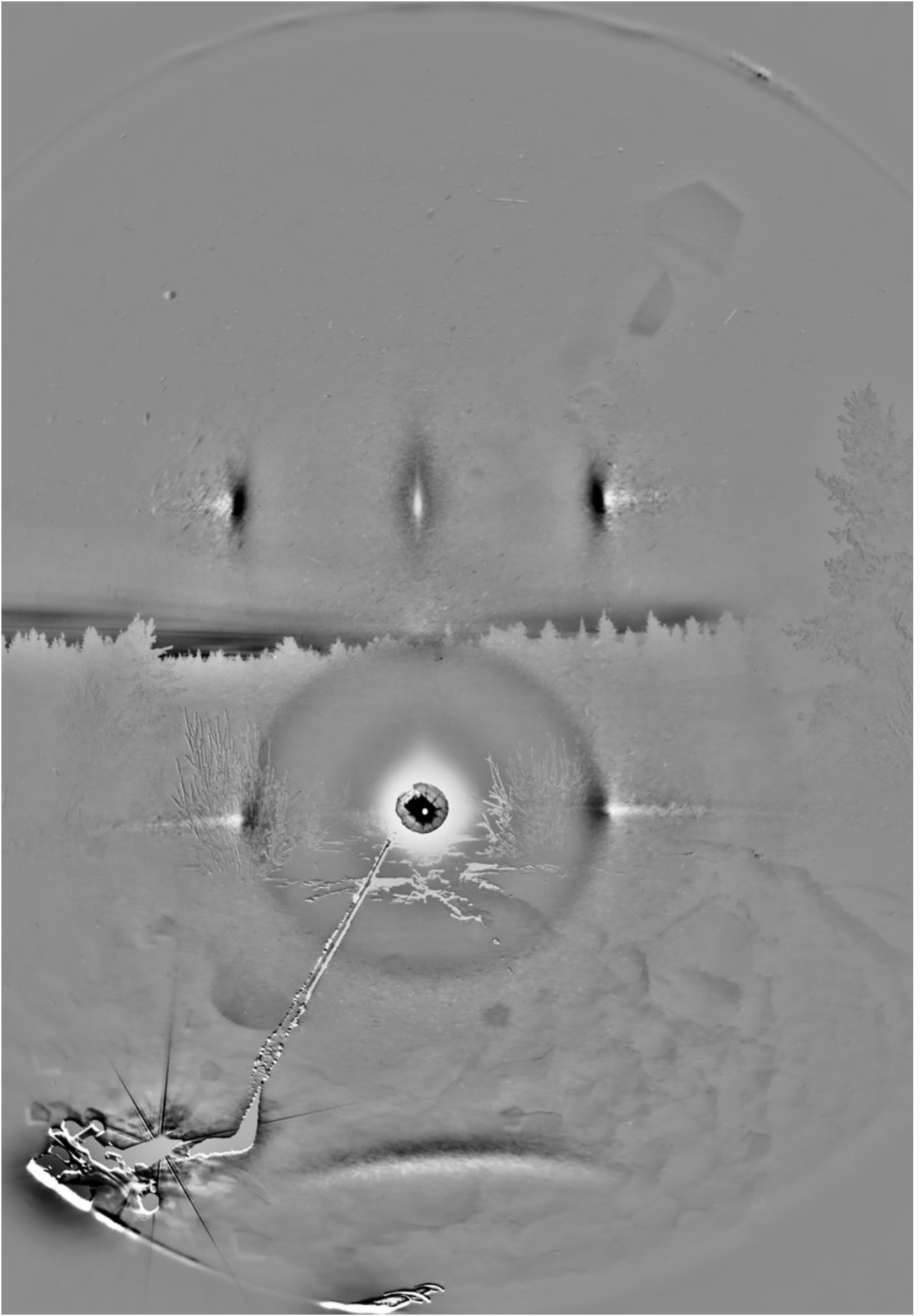
5/21 - 2026-01-23 14:55 UTC

9/10 Jan 2026, Rovaniemi. The odd radius stage was short. Basically just two one minute swarms separated by one minute of only counterparhelia. 4 x 30s.



6/21 - 2026-01-23 15:03 UTC

9/10 Jan 2026, Rovaniemi. There was all the time changes in the swarm. Here the plate boosted for a while. Perhaps I see is suggestion of counterparhelic circle inside the counterparhelia. Max, ave, ave g-r. 25 x 30s



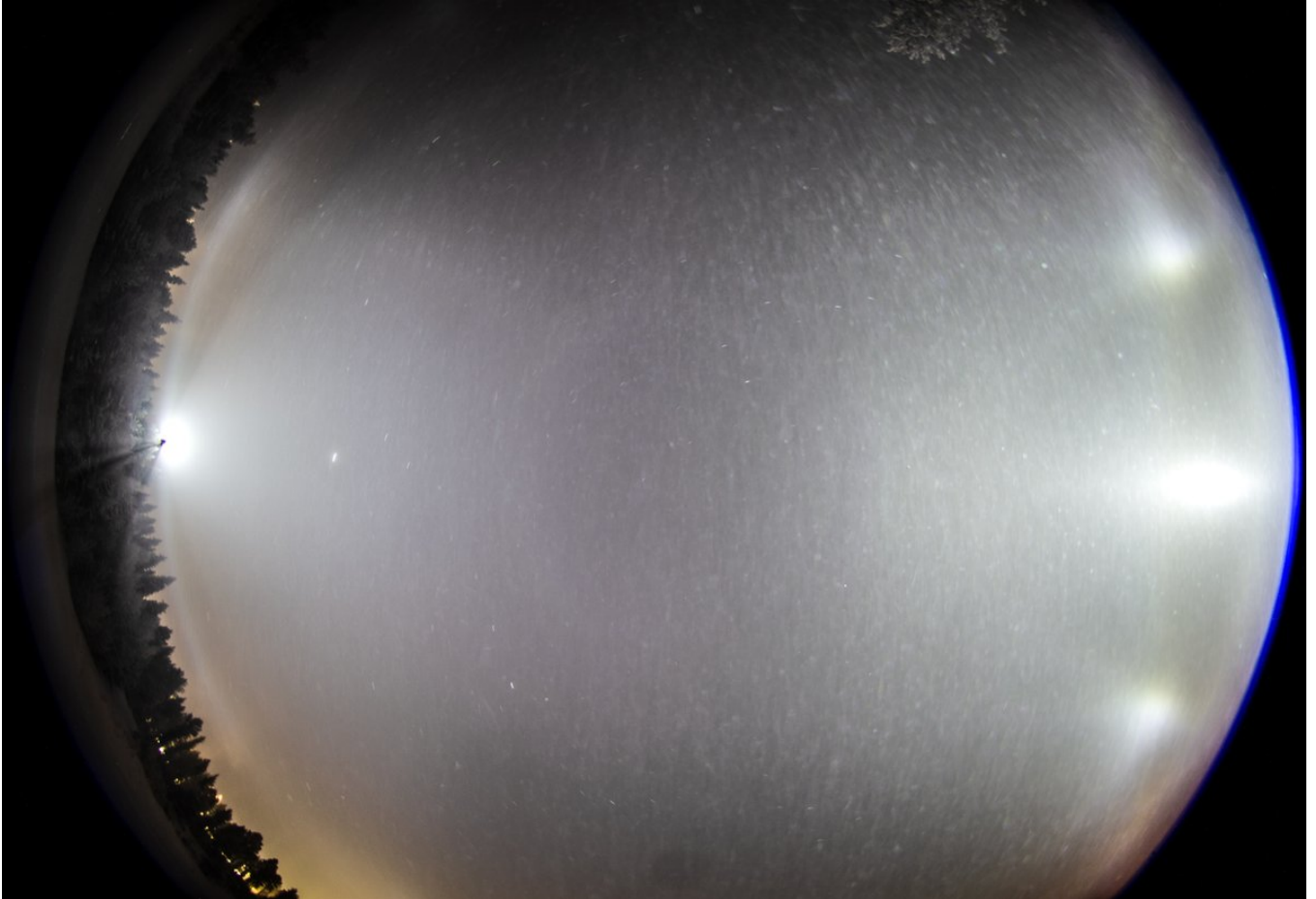


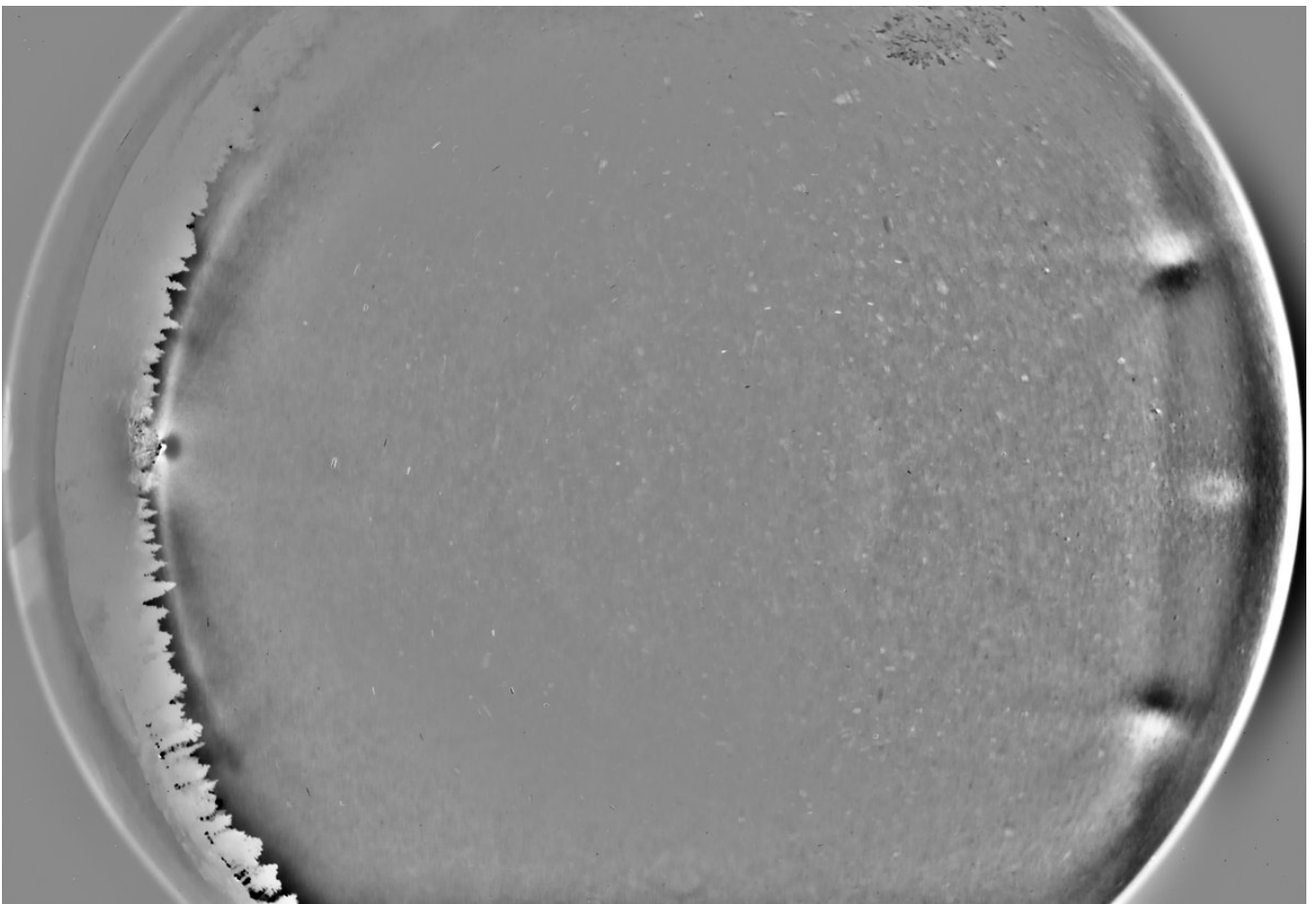
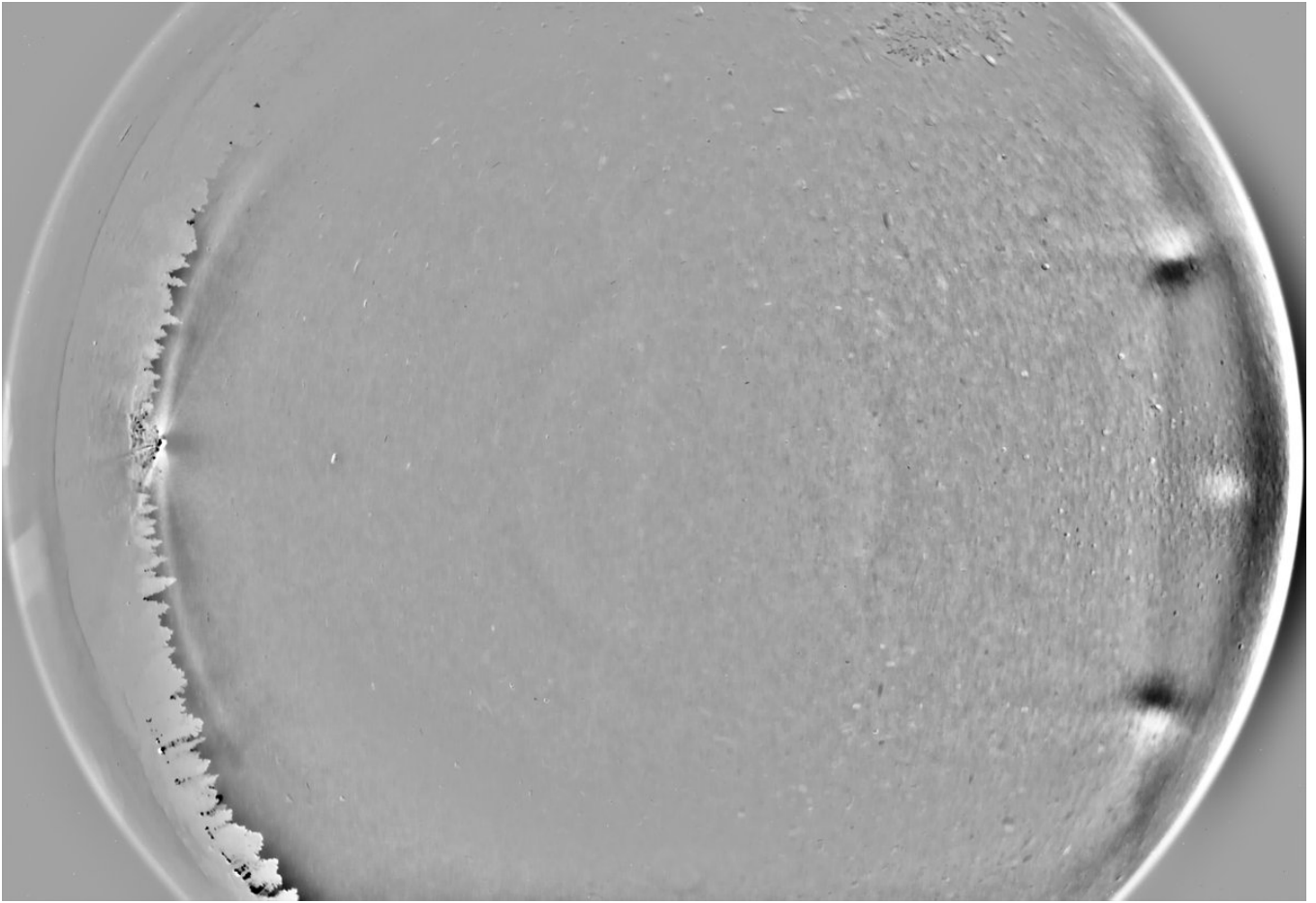


7/21 - 2026-01-23 16:50 UTC

9/10 Jan 2026, Rovaniemi. I should have opted for the darker side-view to cover fully the counter-120° parhelion, which is now visible only partially at the upper edge of the image. Visually it was well fleshed out, but only if you stepped outside the beam. I don't think I have ever seen crystals glitter in counter-120° parhelia – always the fuzzy spots. Why is that? In contrast, in Liljequist counterparhelia you see much glitter.

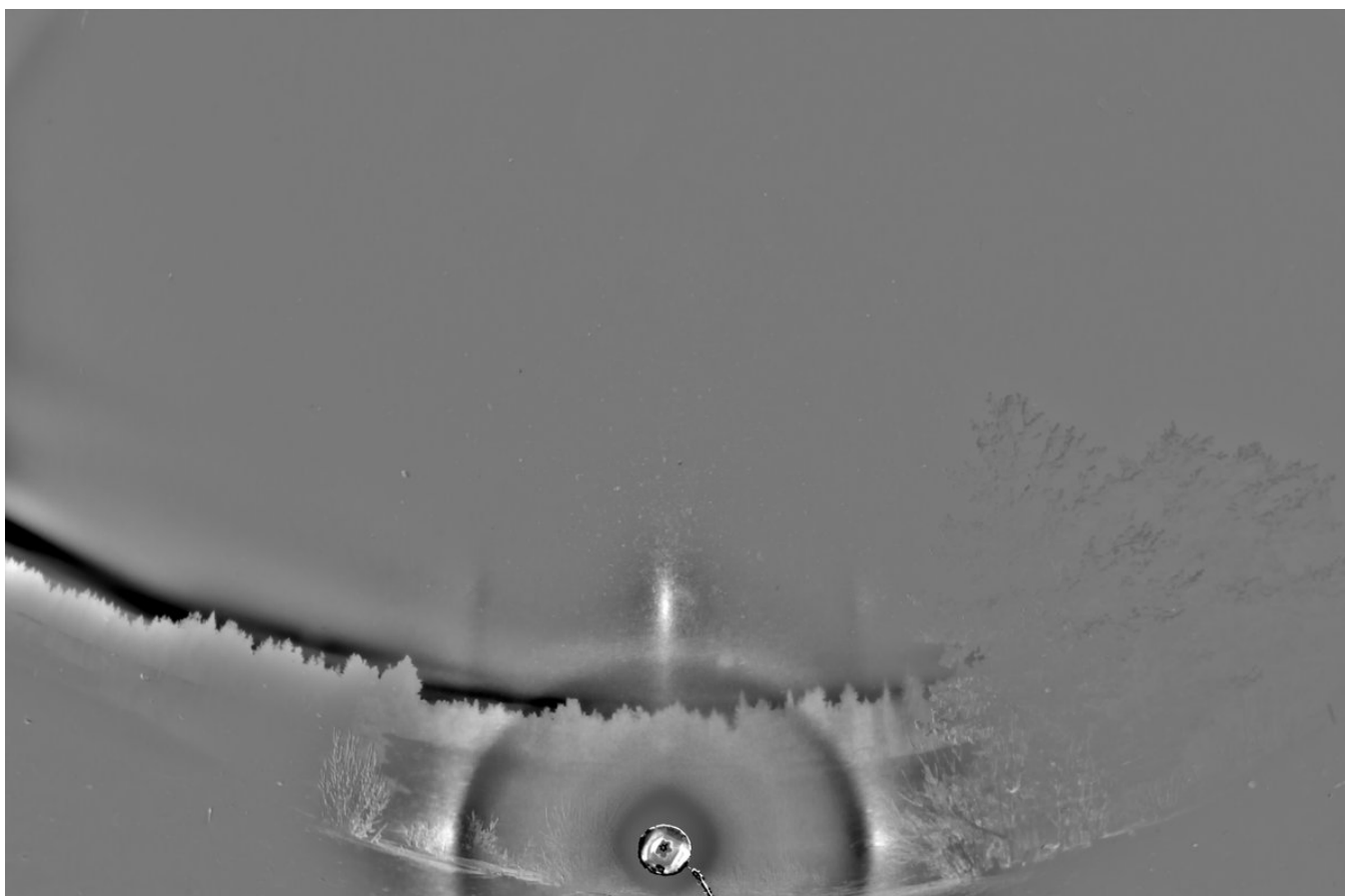
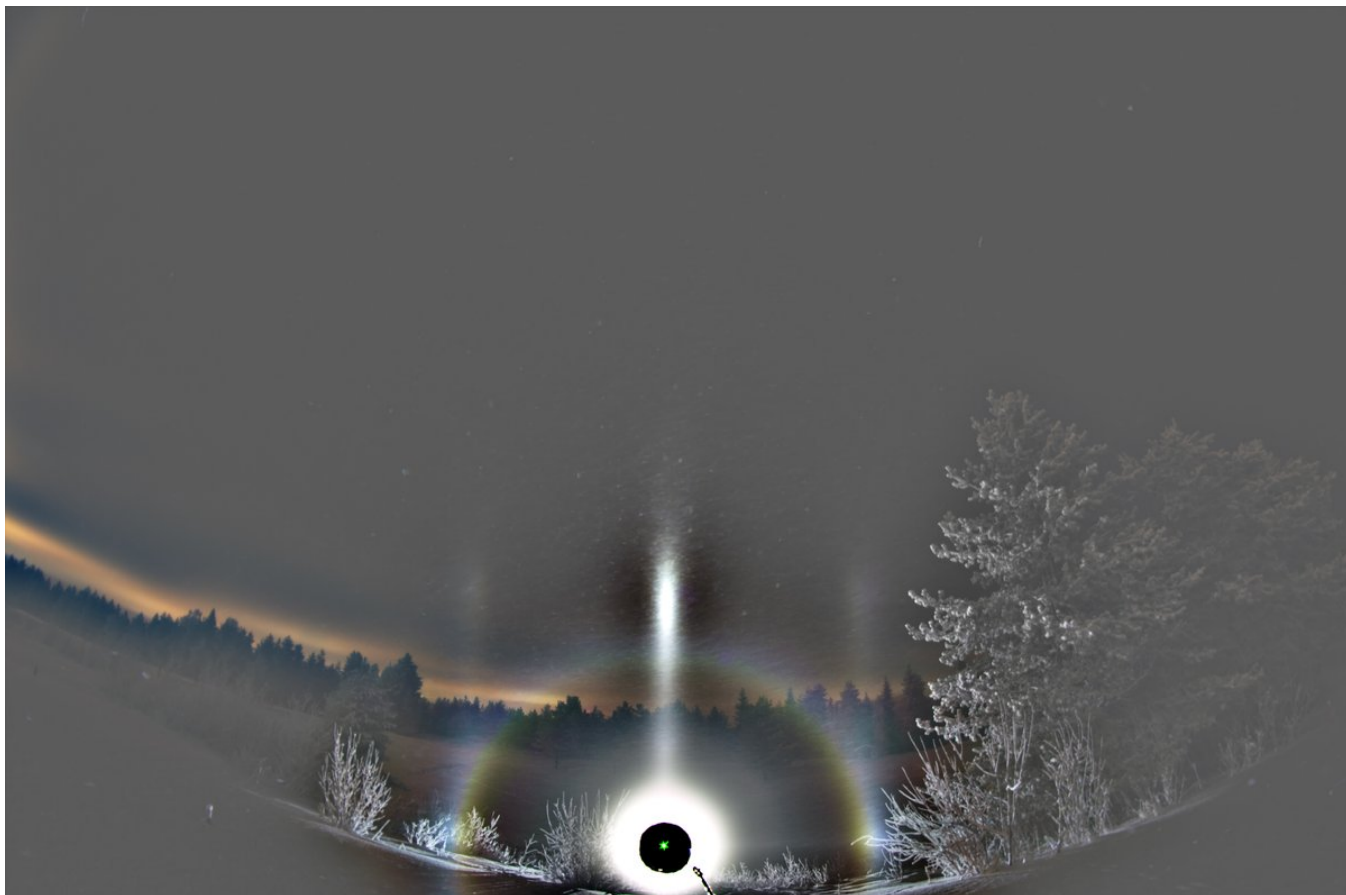
Also weak counternadir arc and counter-nadir Dufay are present. Ave g-r, ave b-g, ave camera raw. 8 x 30s.





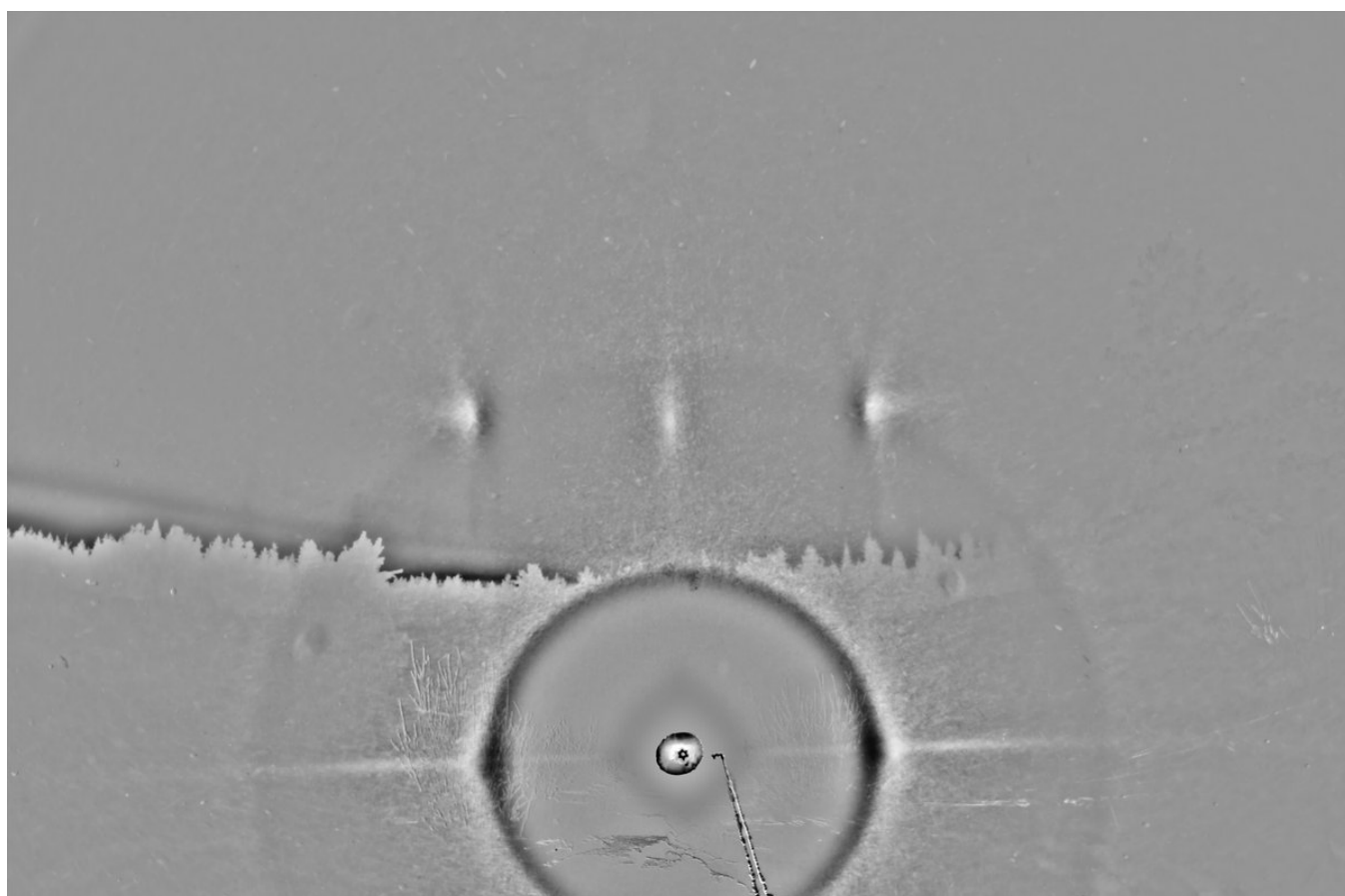
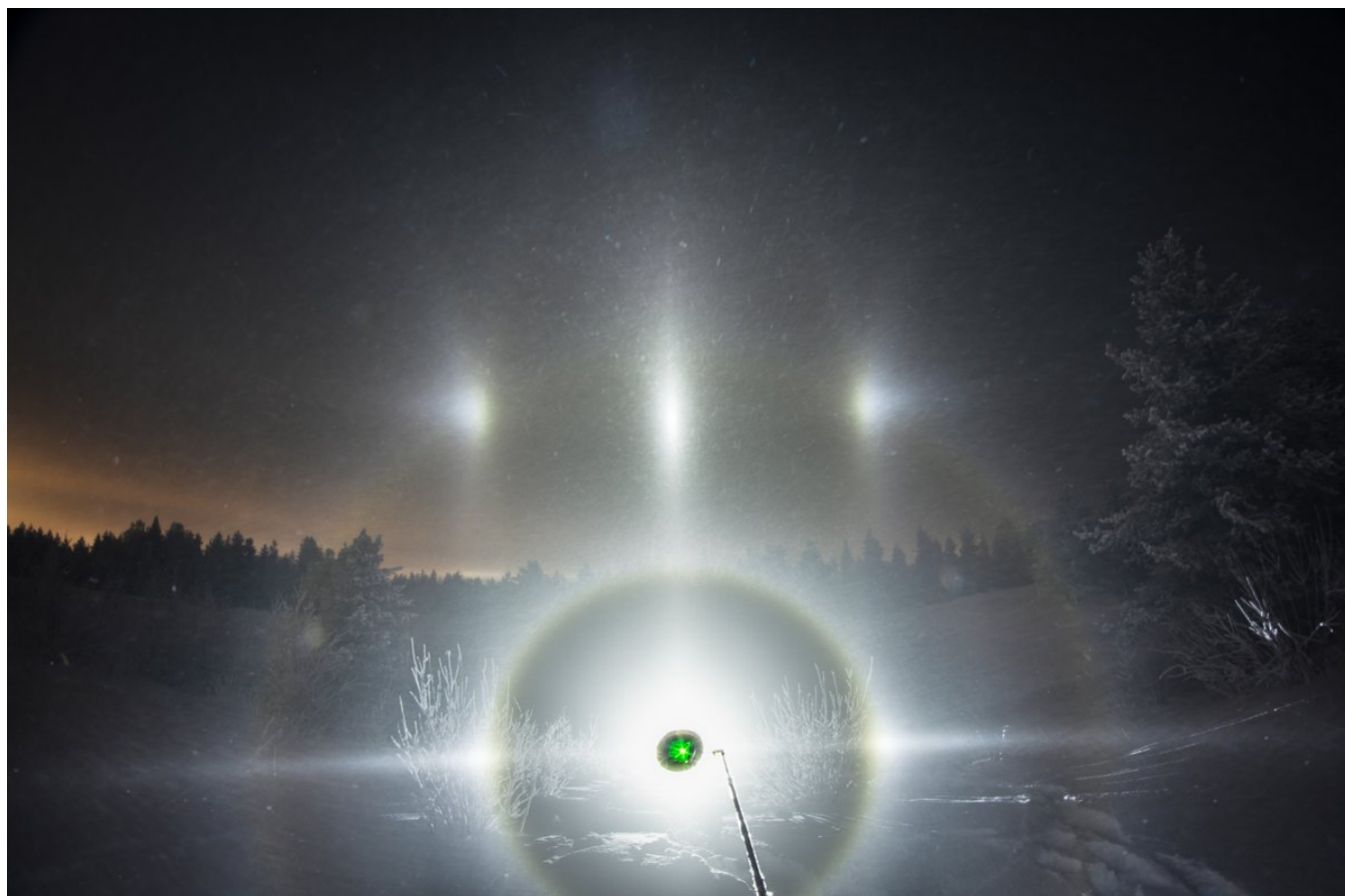
8/21 - 2026-01-23 16:57 UTC

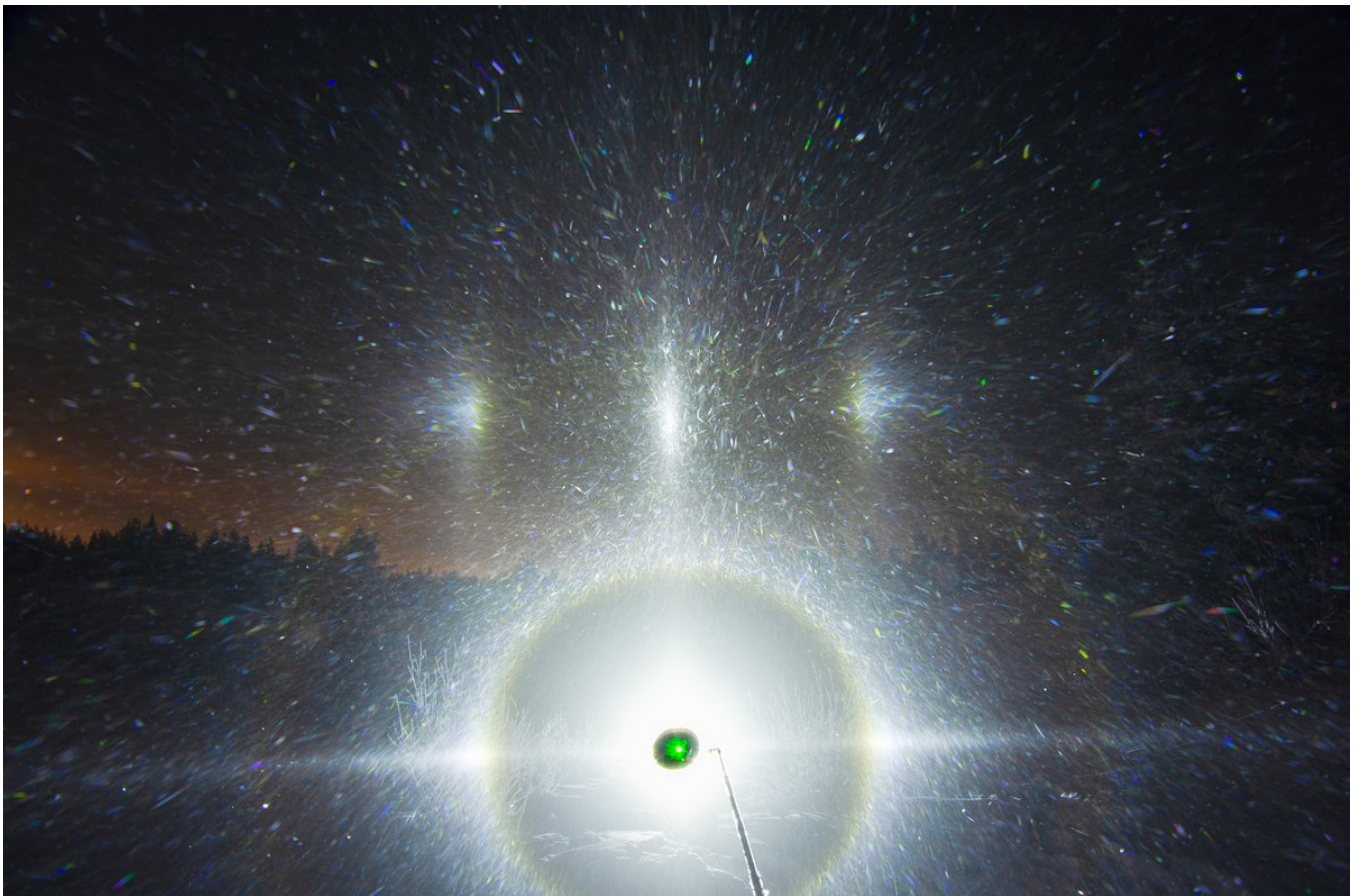
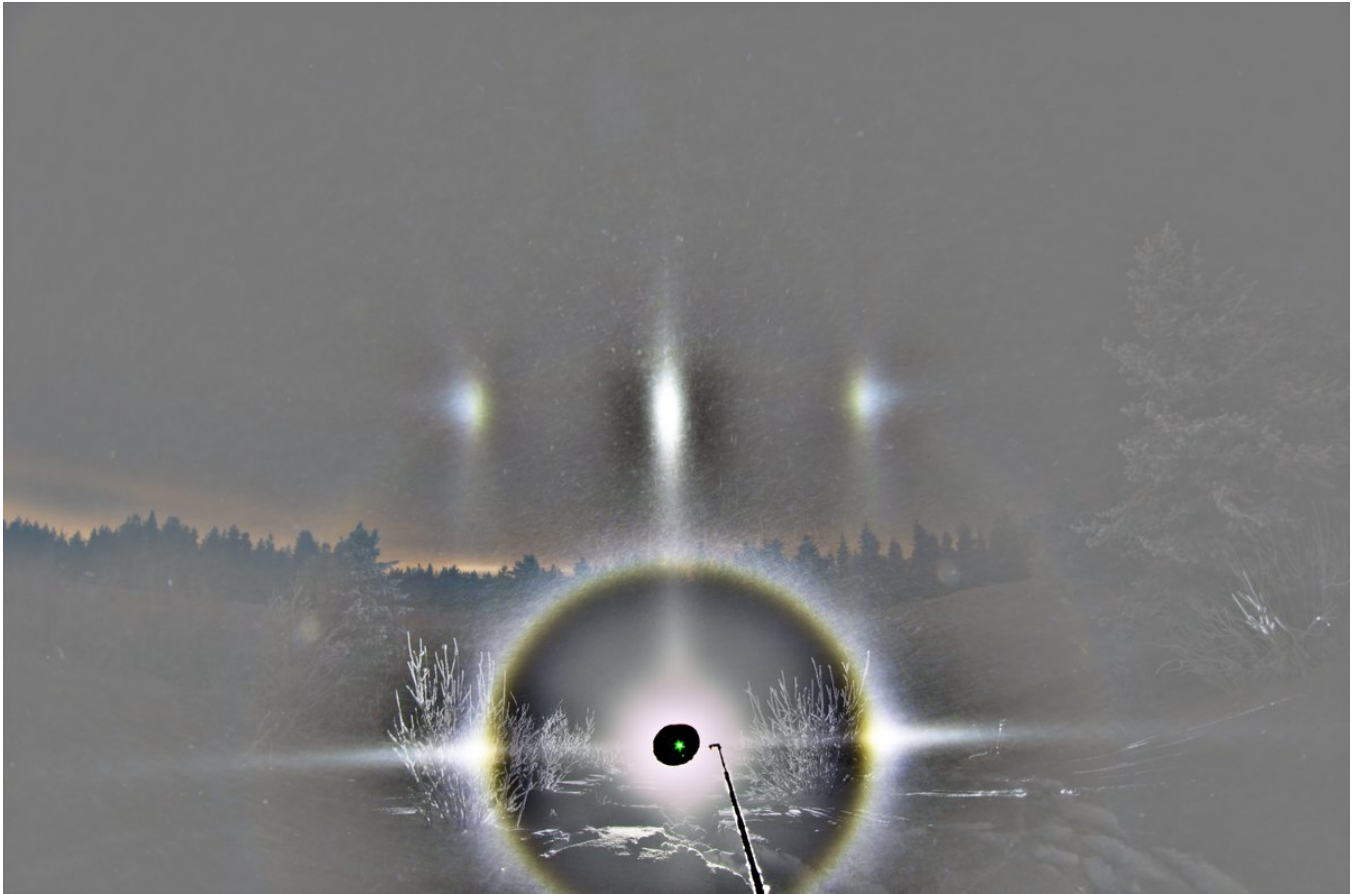
9/10 Jan 2026, Rovaniemi. This was one of the lesser stages, but I publish just to leave a record of the weak Moilanen arc.



9/21 - 2026-01-23 17:21 UTC

9/10 Jan 2026, Rovaniemi. Another view. Around 30 frames in this stack. The last one is max of 4 frames. 22° halo was strong.

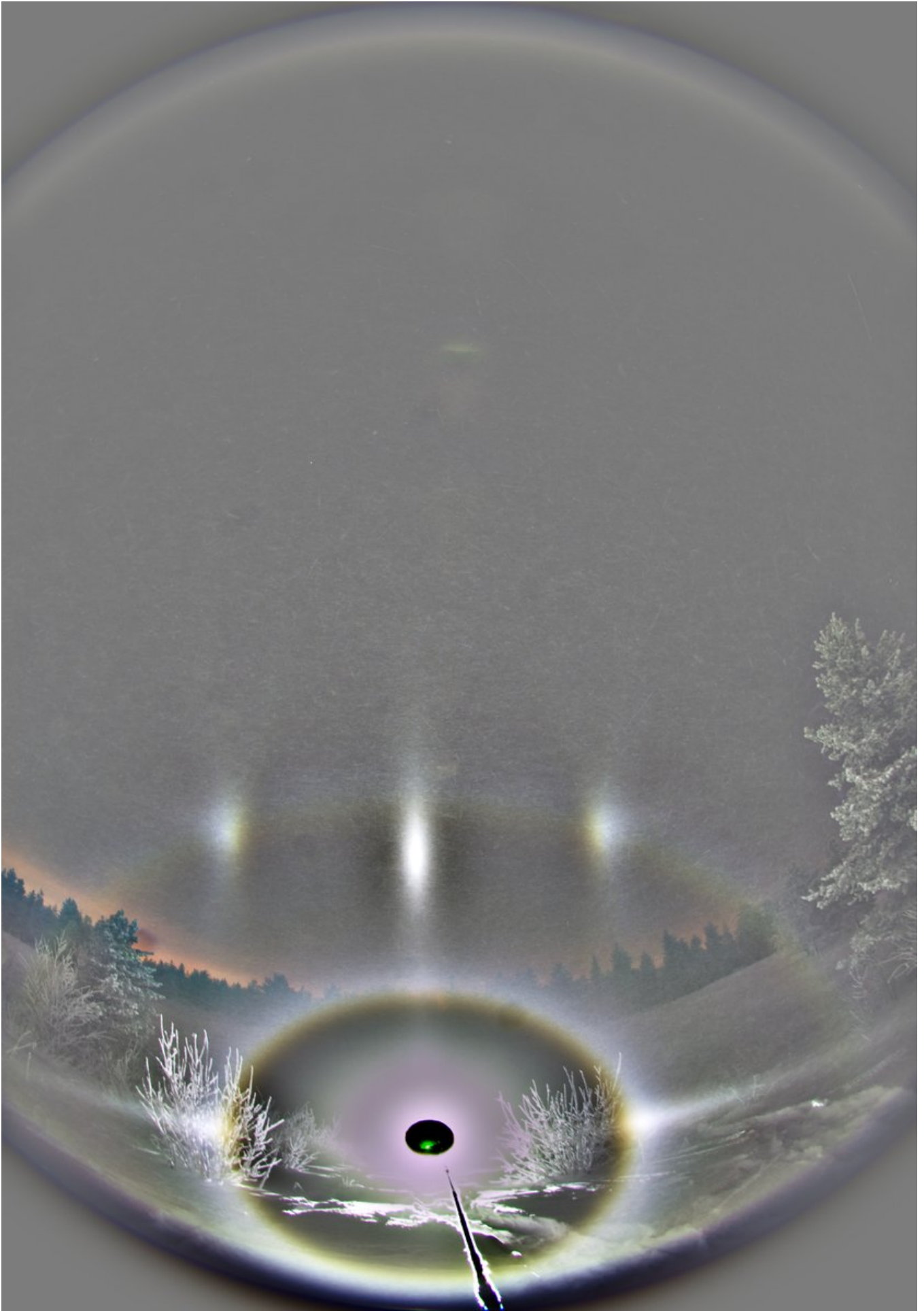




10/21 - 2026-01-23 17:24 UTC

9/10 Jan 2026, Rovaniemi. 54 x 30s. The last max stack has smaller number of frames.



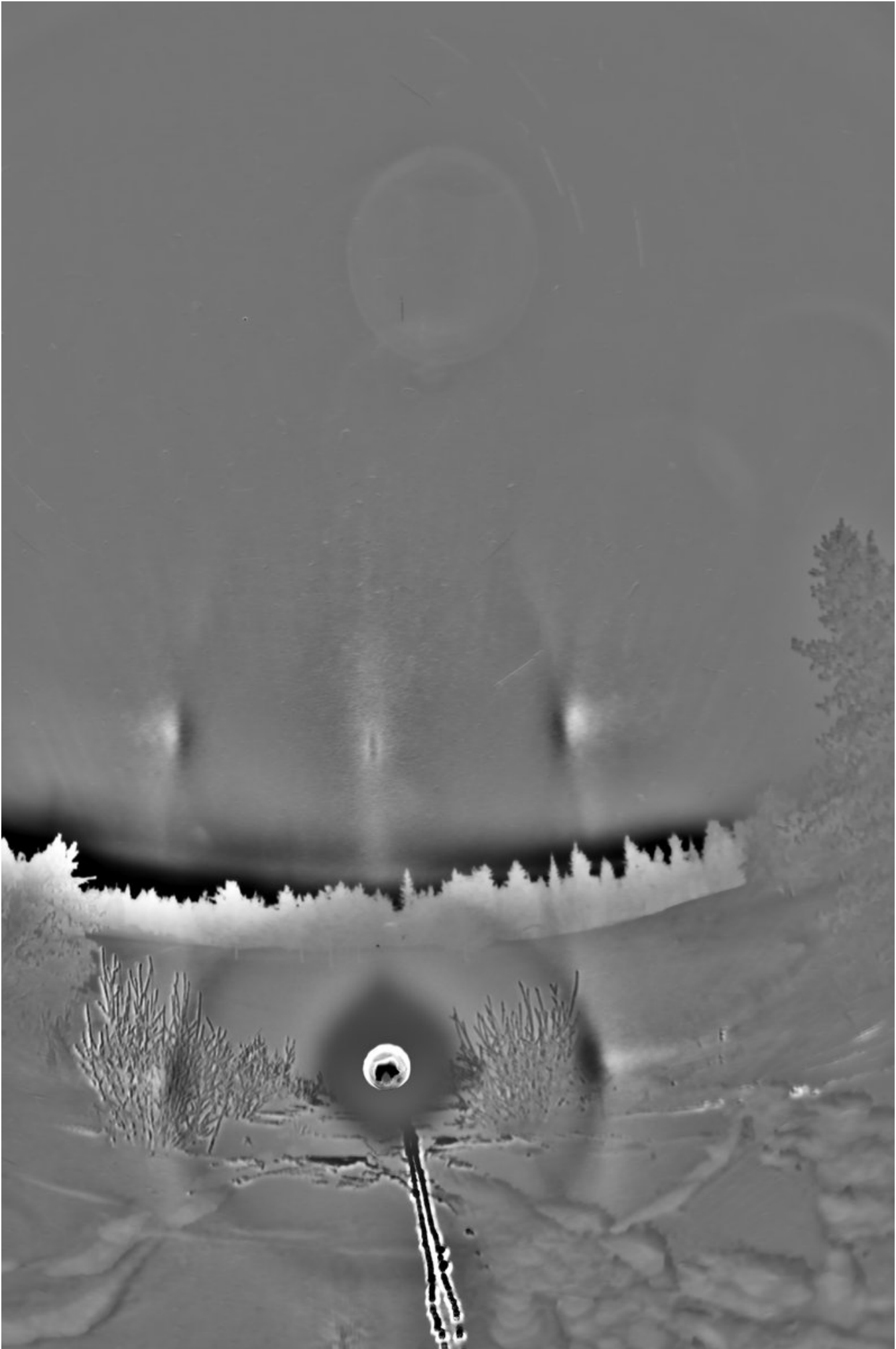






11/21 - 2026-01-23 17:28 UTC

9/10 Jan 2026, Rovaniemi. 63 x 30s.



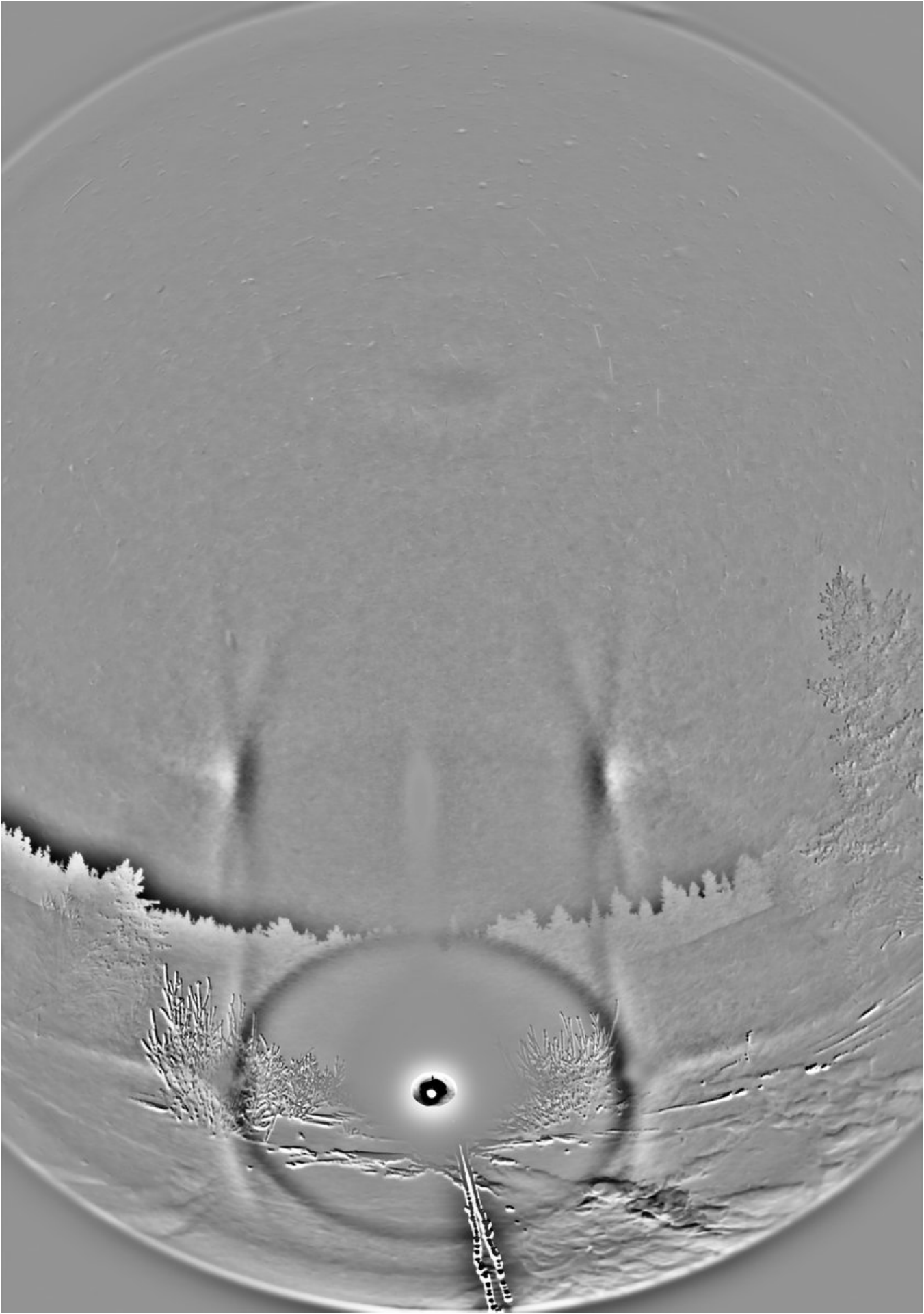


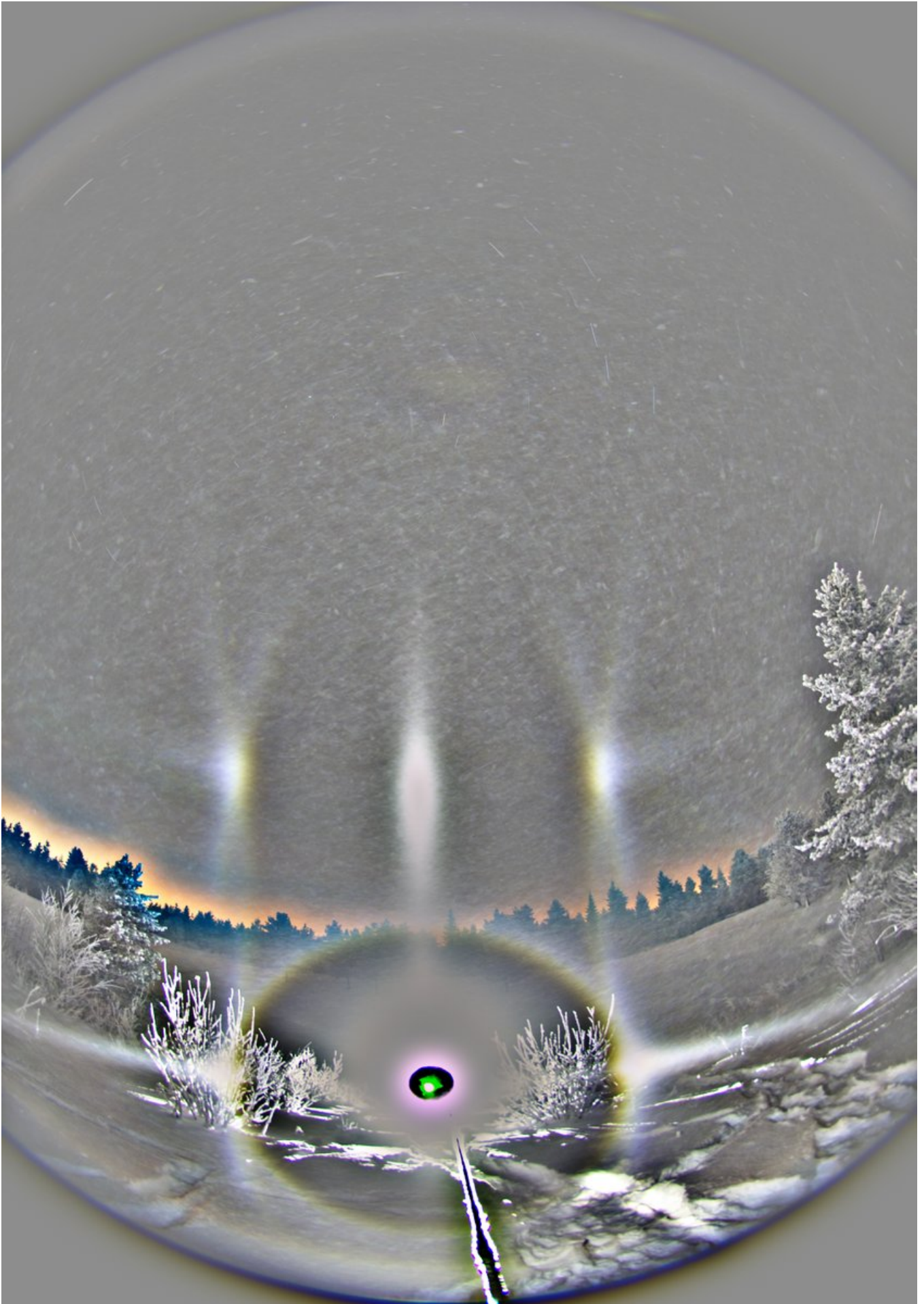
12/21 - 2026-01-23 18:33 UTC

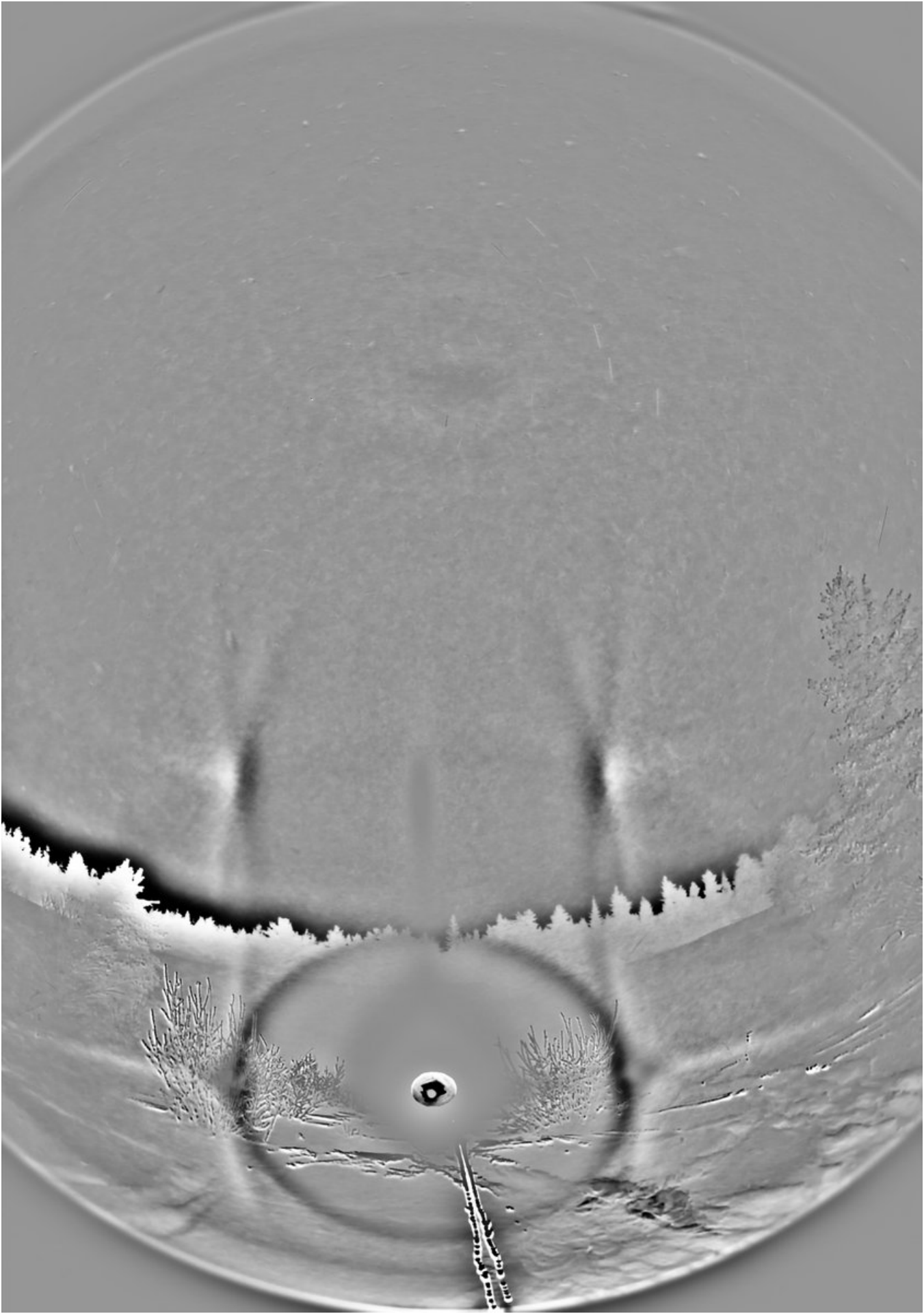
9/10 Jan 2026, Rovaniemi. 37 x 30s. If your focus is on the Liljequist arcs (which is maybe the best thing to do in plate displays), this shows the importance of covering also well below horizon for the completeness of the picture. So are those arcs below parhelia circular Lowitz or concave Liljequist?

We need to get this kind of display in a dark location. And if the lamp is enough low to separate the parhelia clearly from 22° halo, we get also better idea of the possible Lowitz action.

I don't know if the fact that no sharp Liljequist arc is visible even merits a note. It really does not seem to exist.

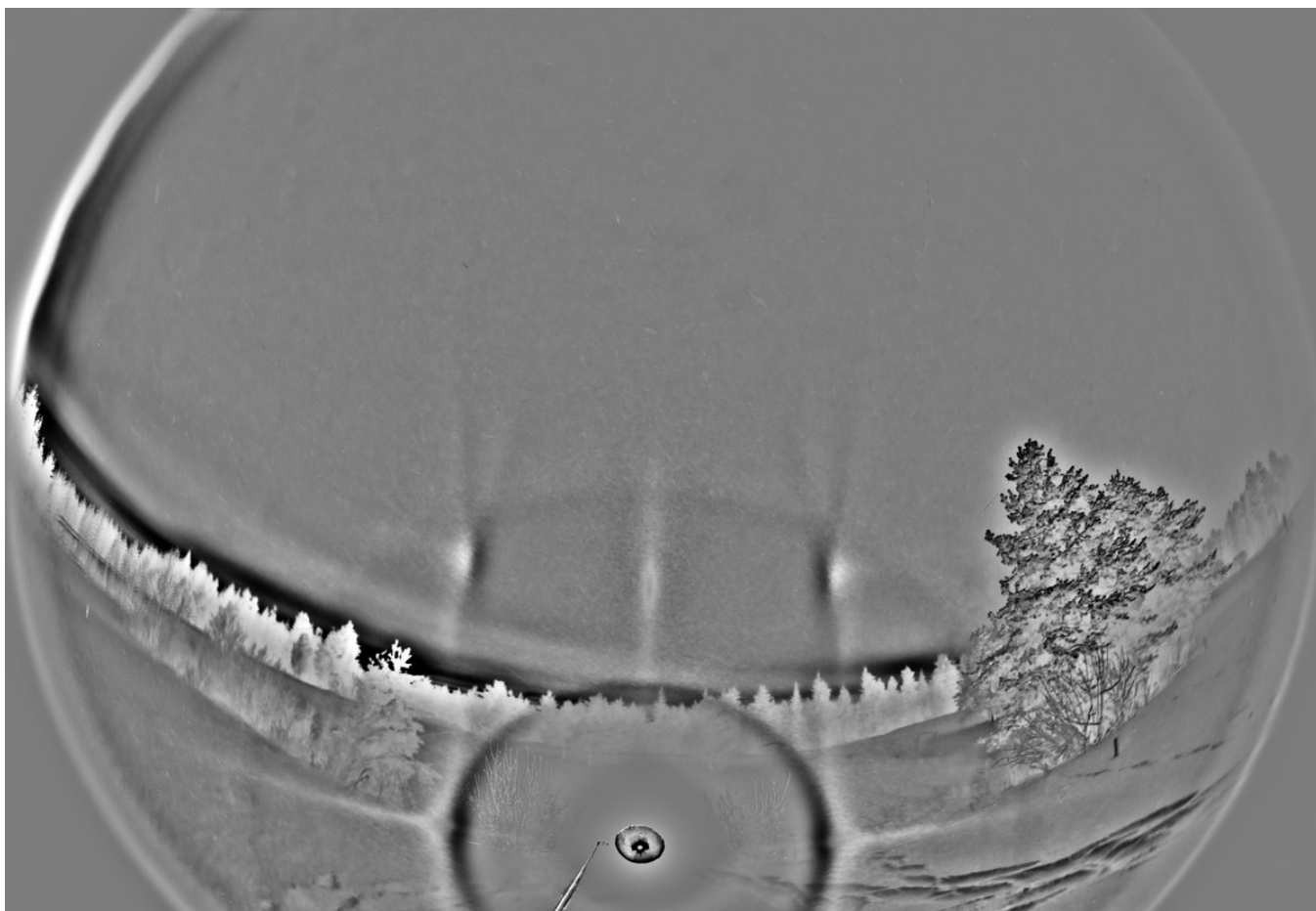


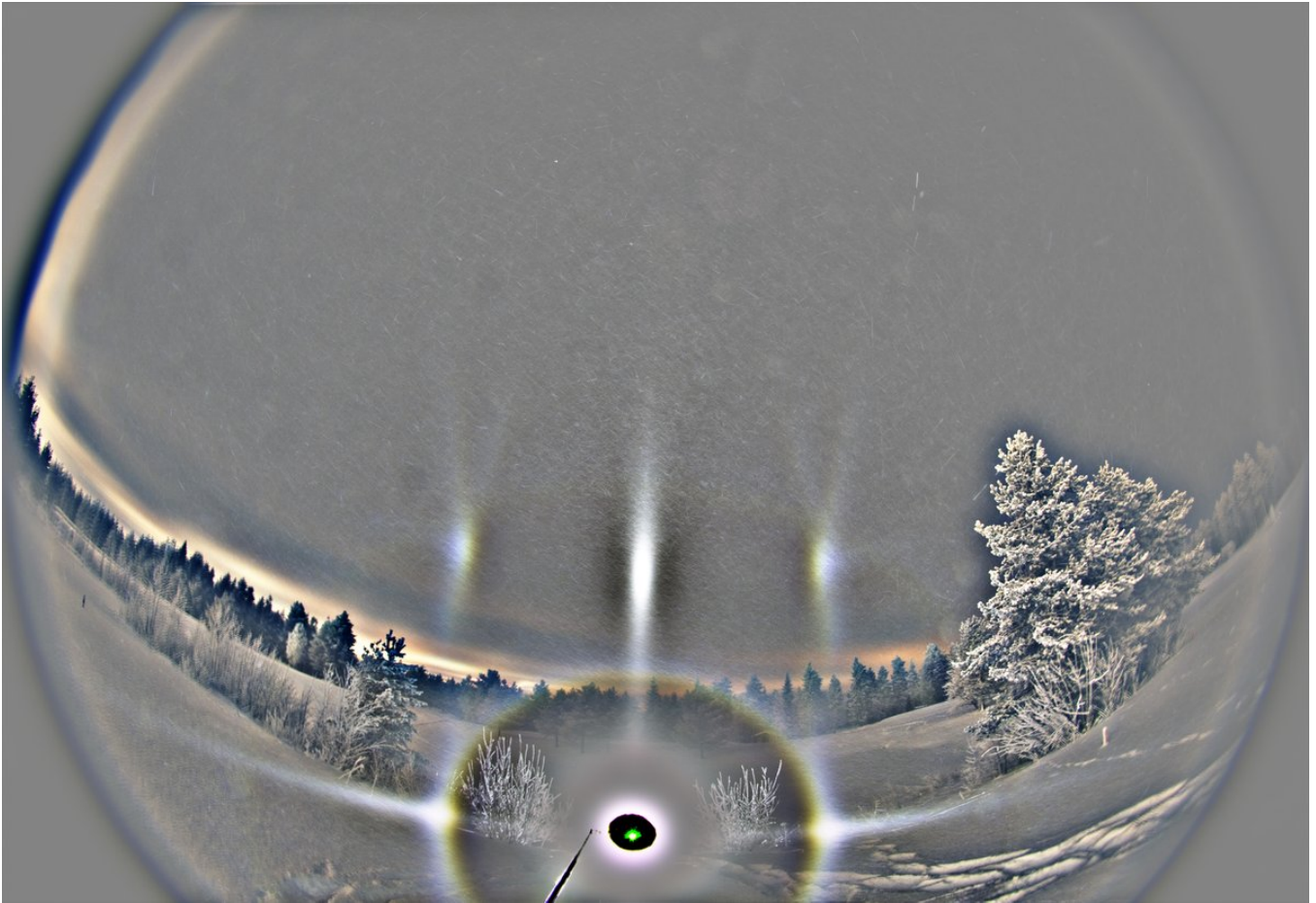




13/21 - 2026-01-23 18:40 UTC

9/10 Jan 2026, Rovaniemi. This seems to be the final set. 40 x 30s.





14/21 - 2026-01-26 00:02 UTC

9/10 Jan 2026, Rovaniemi. I found more photos around the odd radius stage when browsing the memory card (lucky I had not formatted it yet) and made a longer stack of 11 frames (8172, 8182-8191). The gif here is flashing this stage with the preceding "normal" stage.

What strikes immediately from this comparison is that Liljequist arcs turn spot-like. If I had only this stack, not the earlier stage, I would have checked both the upper and lower 24° plate arcs and possibly gone wrong. I think we should be even more careful now with the identification of the "ten and two o'clock spots" spots which are not outright clear 24° spots, particularly in displays that have Lowitz / Liljequist arcs. There have accumulated quite many of these mysterious spots over the years. One is the display by Martikainen on 6 February 2021 in Juva: <https://t.co/OHjax3YOwW> Another one I can remember right off the bat is the later stages of the Jutland display: <https://t.co/Vckjx7dj8p>

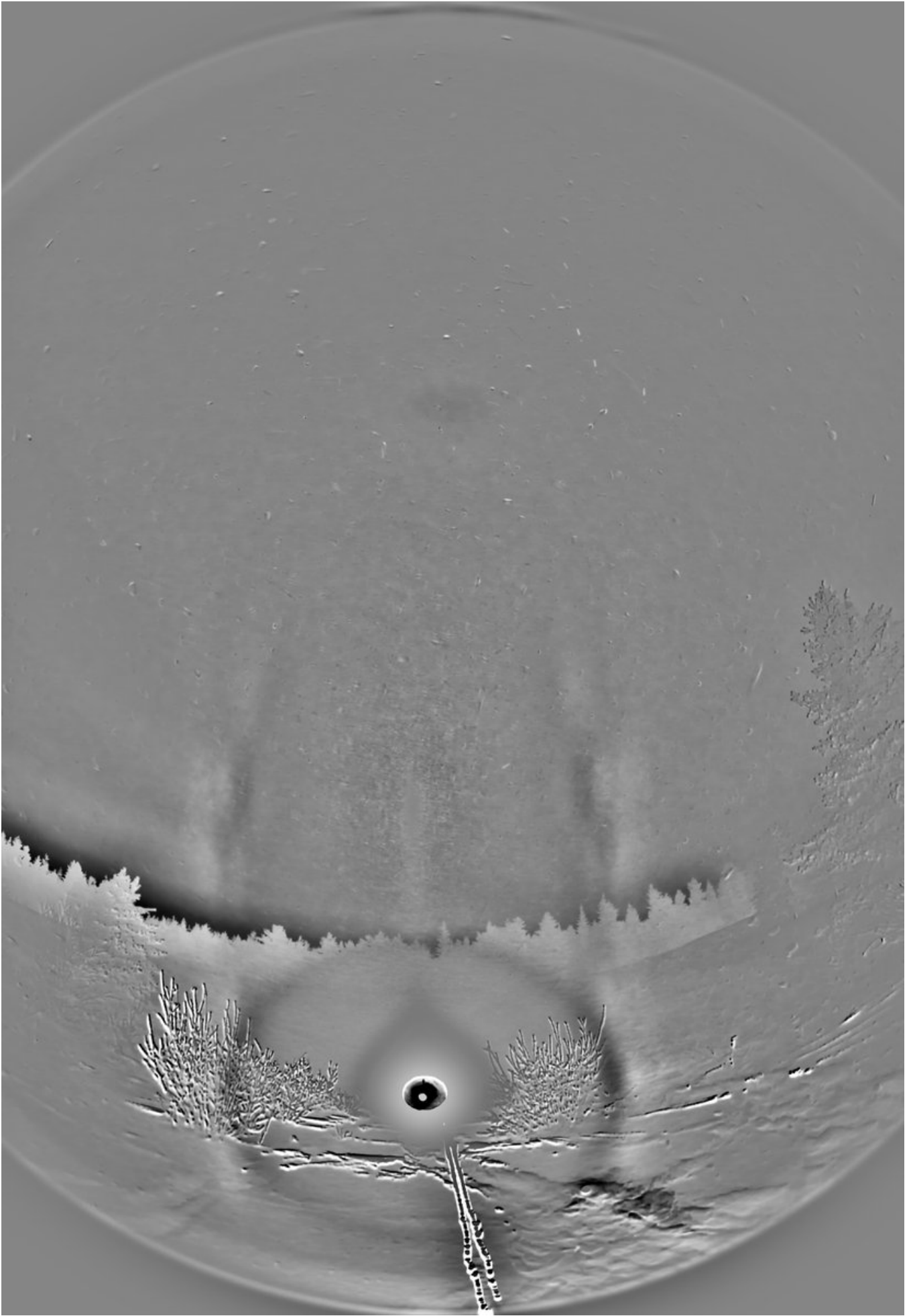
But that's not all. Above the treeline is seen a more profound change in the concave Liljequist arcs as they become tilted outwards. Something also appears outside the counterparhelia, maybe just a continuation of these tilted arcs, maybe a separate feature.

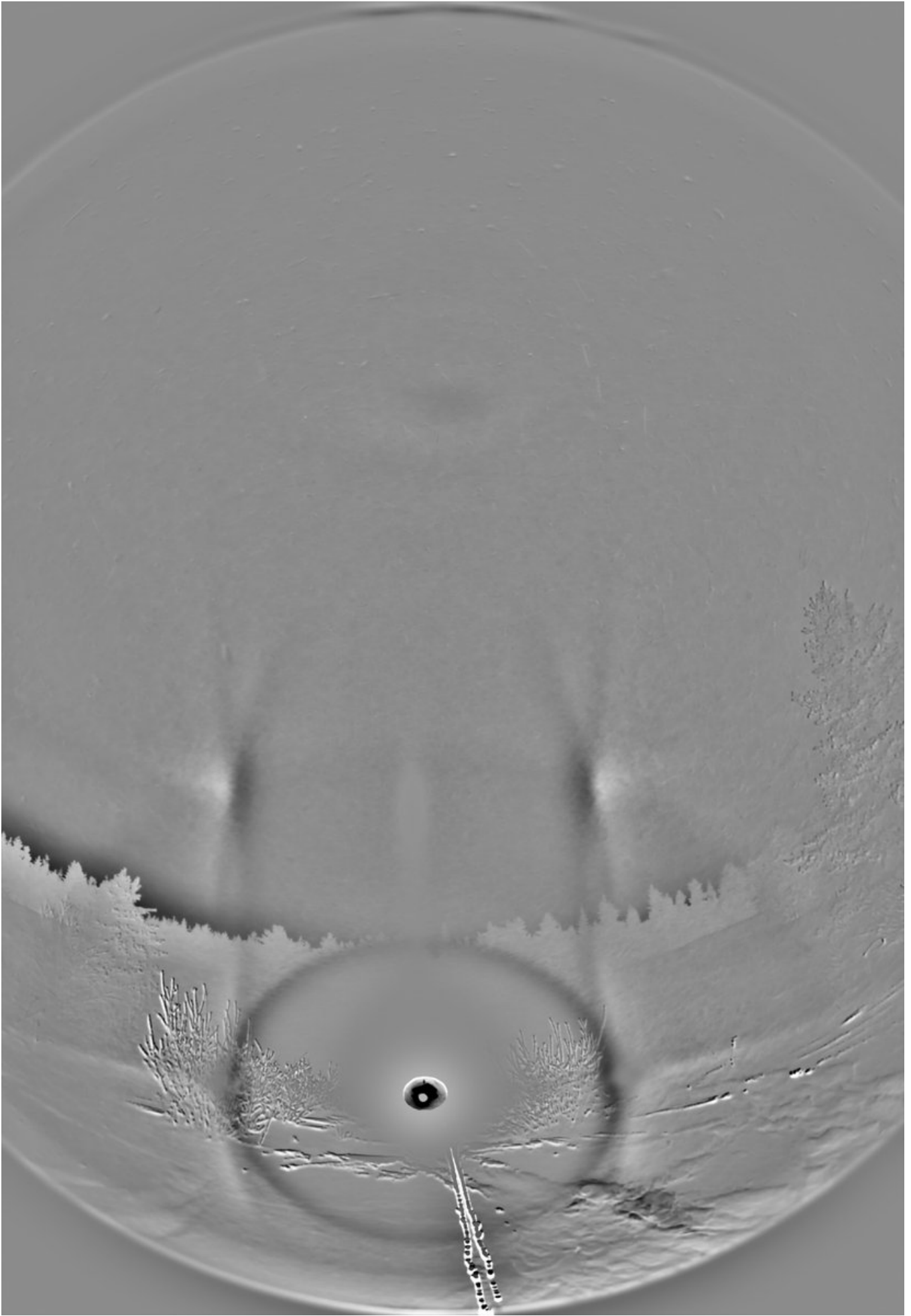
There is familiarity to this stuff. We have had similar kind of scenes in earlier displays: <https://t.co/nhNBZKHivK>

[video: 2015576089821950157-G_jDv2tWMAAIQYI.mp4]

15/21 - 2026-01-26 00:12 UTC

9/10 Jan 2026, Rovaniemi. Three versions of the 11 frame stack and the earlier stage.





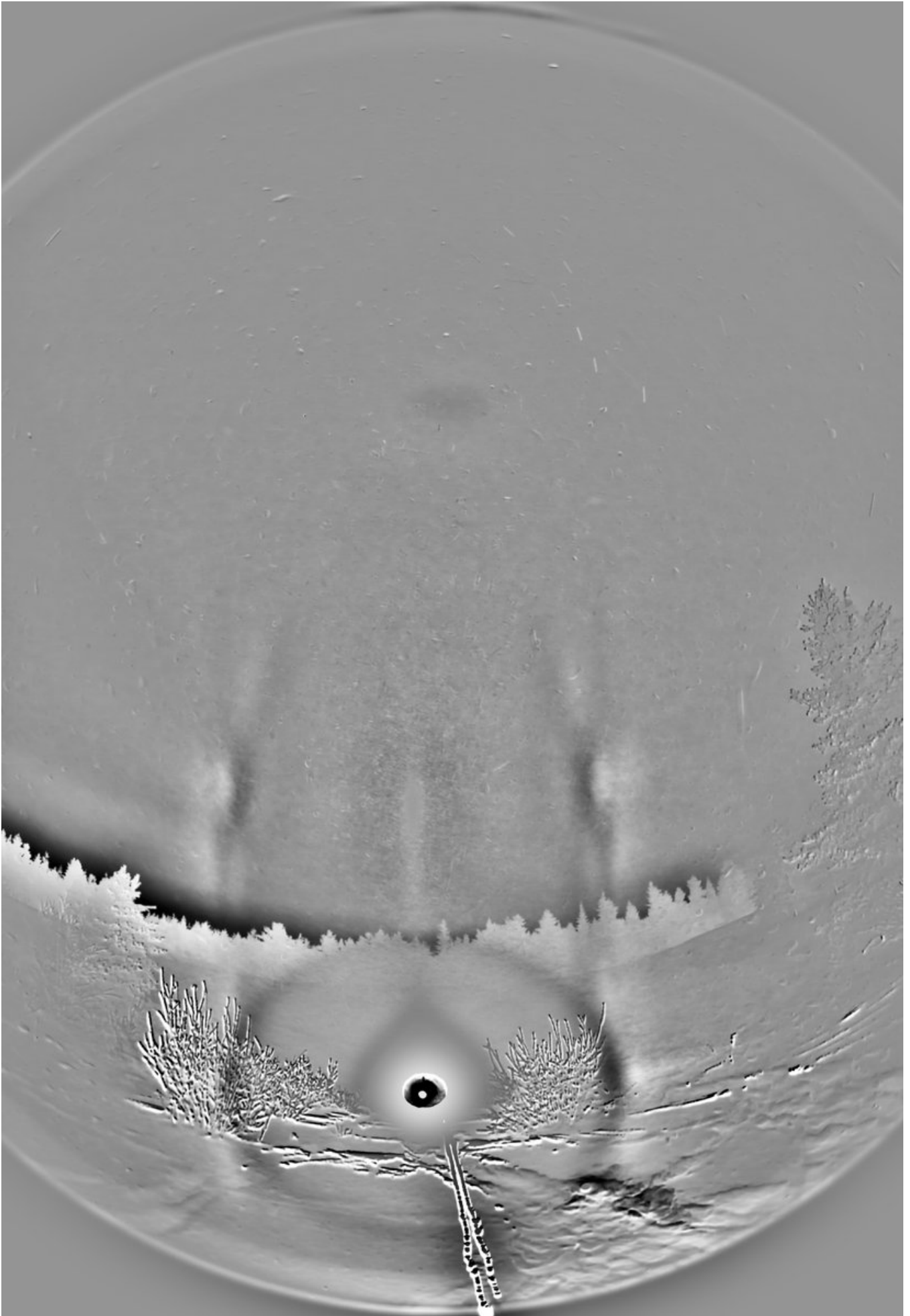




16/21 - 2026-01-26 00:37 UTC

9/10 Jan 2026, Rovaniemi. It was actually possible to stack a longer sequence with an improvement. 22 x 30 s, 8170-8191. This stack (and also the one above) includes the 4 frames I used earlier in this thread to show the counter-18° plate arcs, but it appears that the extra frames added eat them away. Well, maybe there is a suggestion still there.

About the arcs shooting past outside the counterparhelia, another display where this feature is clearly fleshed out is the one on the night of 28/29 November 2016 in Rovaniemi: <https://t.co/w4MZsmHixT> But it has none whatsoever odd radii.



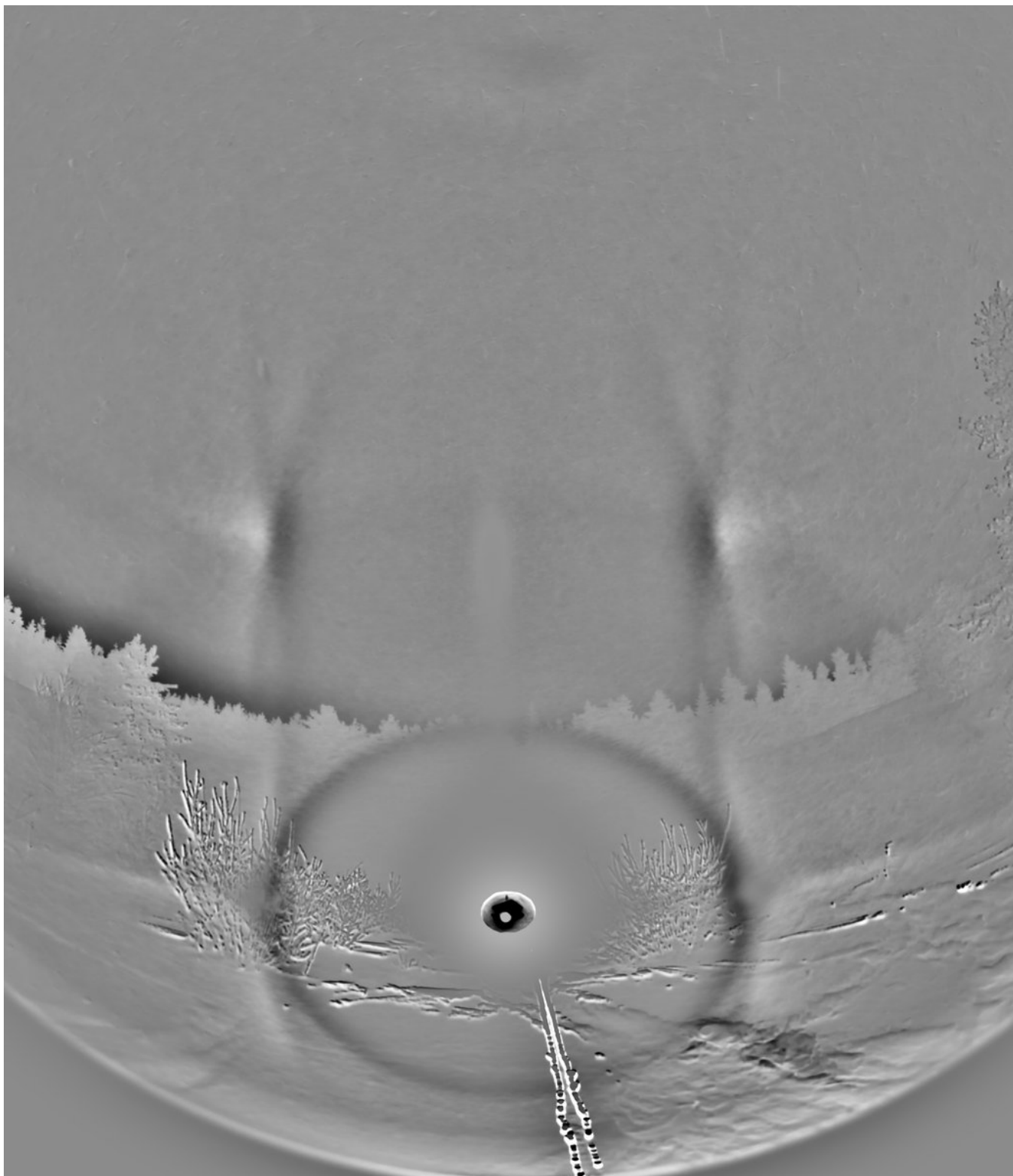
17/21 - 2026-01-26 11:04 UTC

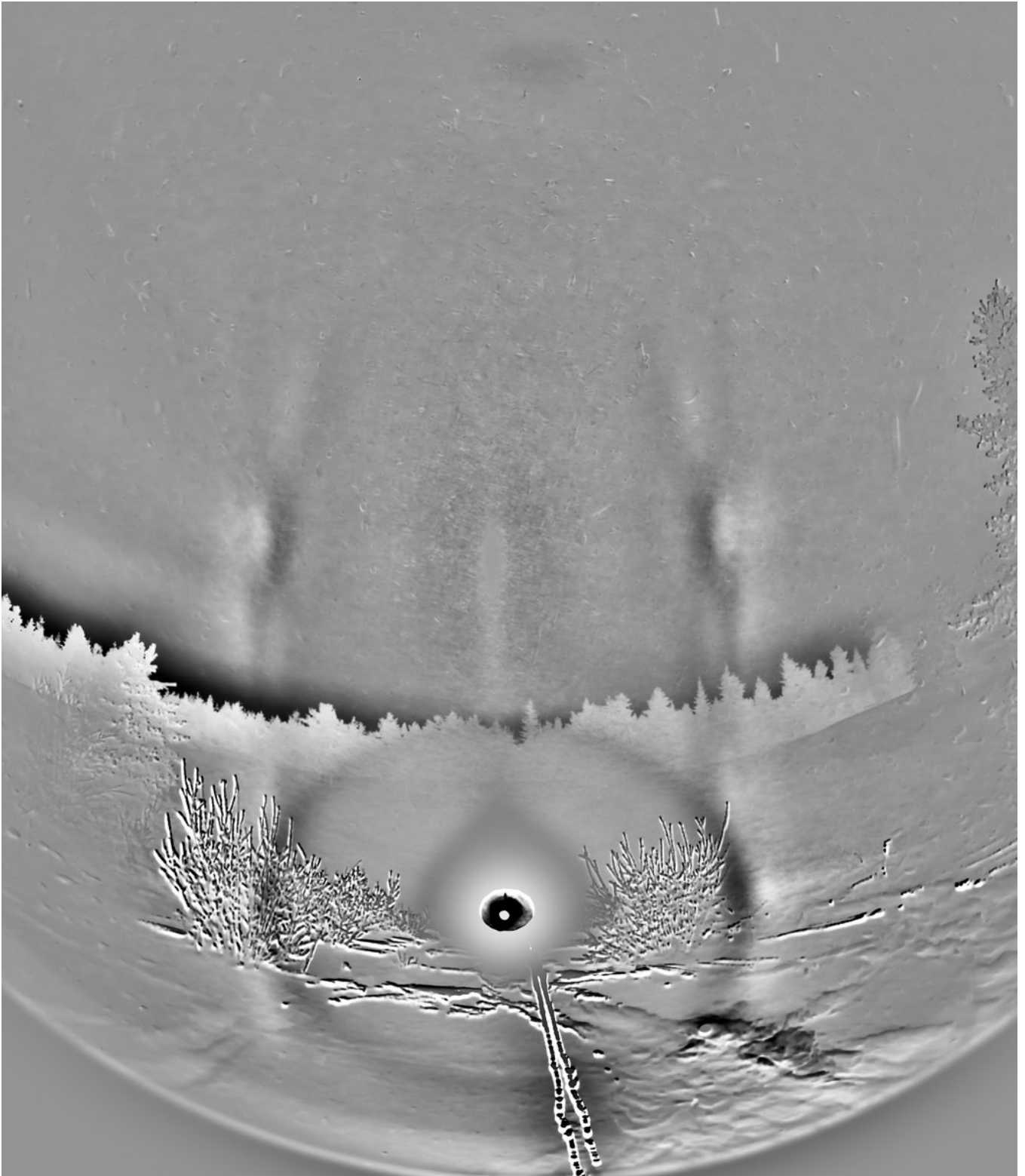
9/10 Jan 2026, Rovaniemi. Flashing the longer 22 frame stack with the earlier stage.

[video: 2015742659575288042-G_lbCioWsAEsBBx.mp4]

18/21 - 2026-01-26 11:07 UTC

9/10 Jan 2026, Rovaniemi. The images of the previous gif.



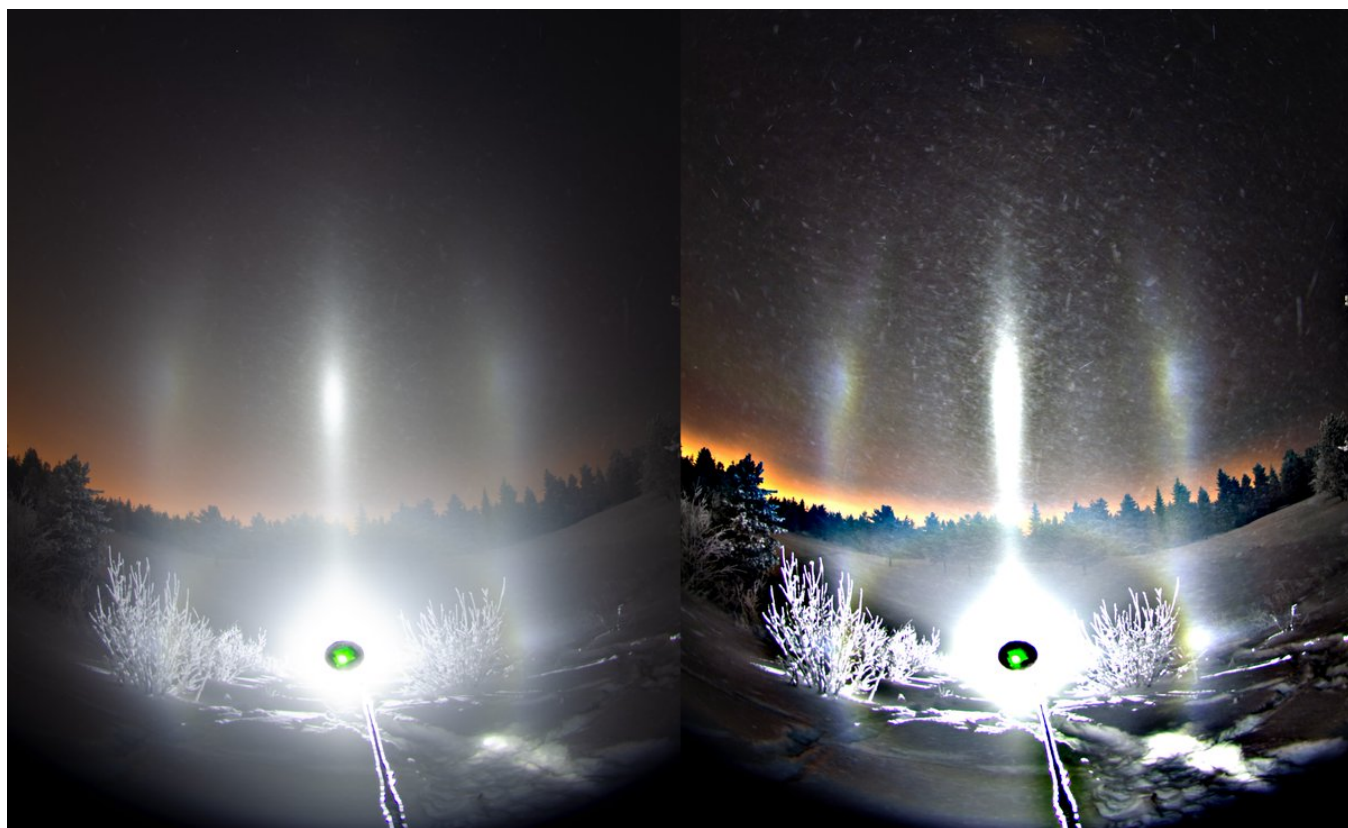


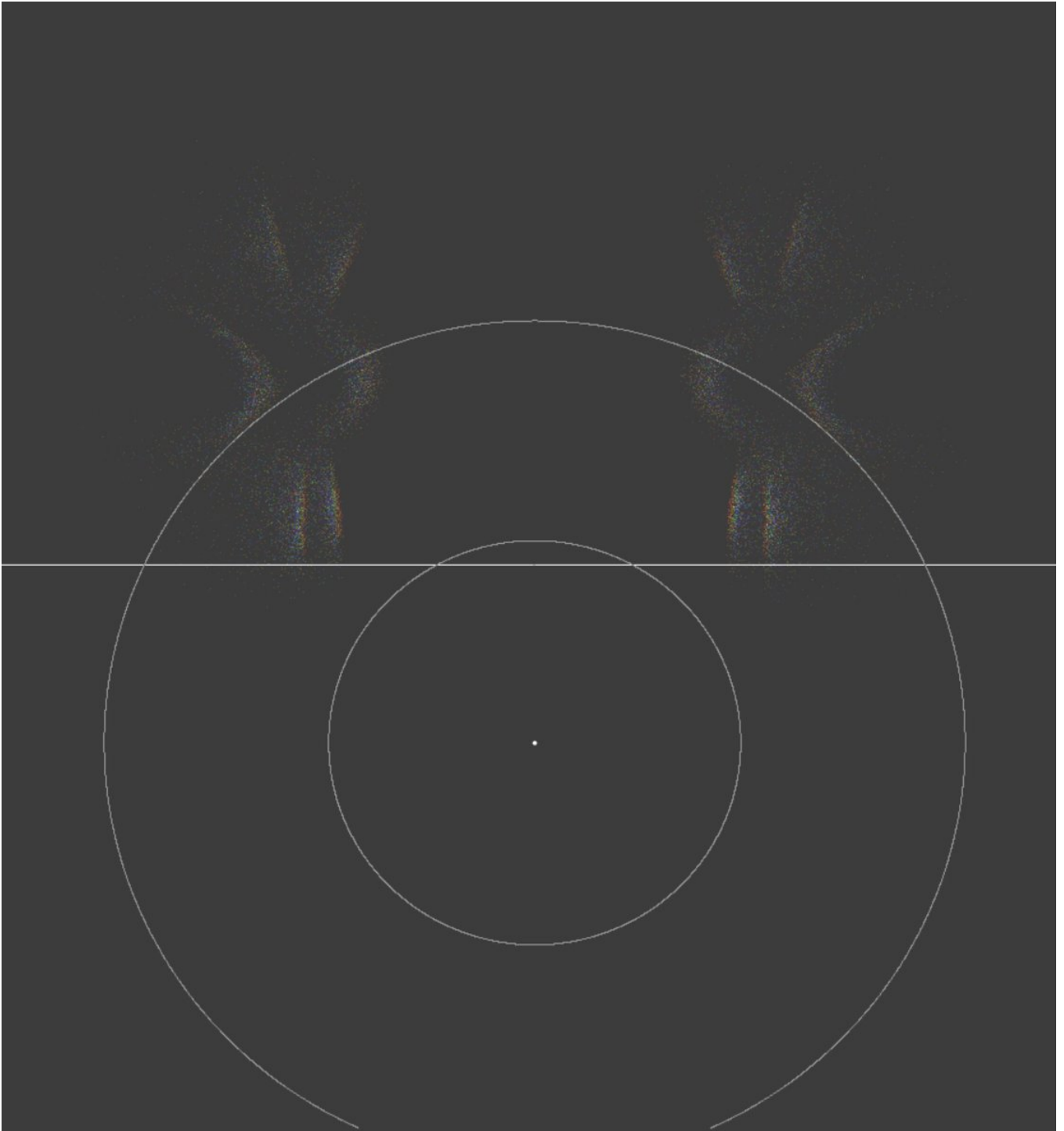
19/21 - 2026-01-26 12:17 UTC

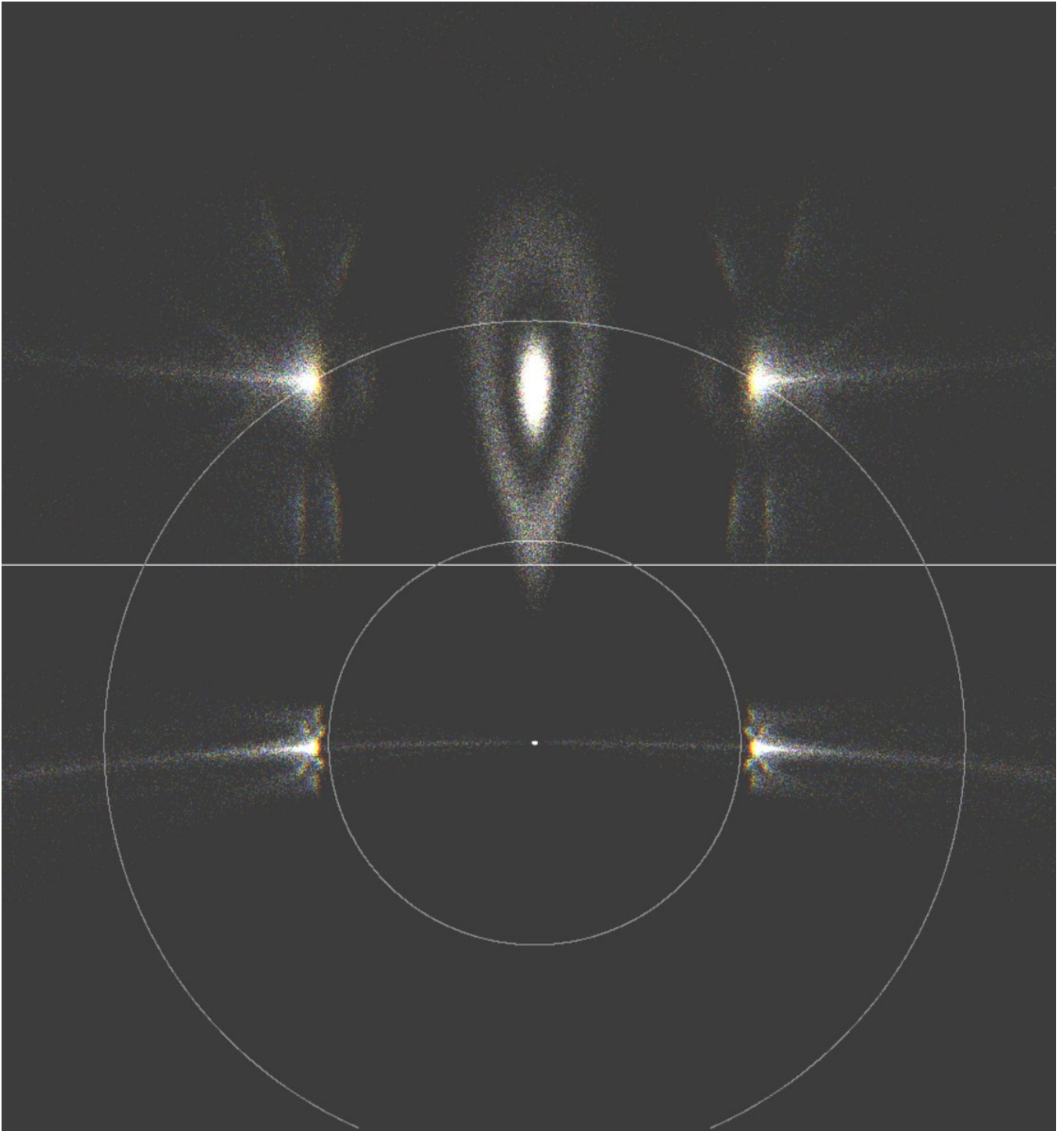
9/10 Jan 2026, Rovaniemi. Attempting simulation. Lowitz crystals tilted 8 degrees, small deviation of 1.5 degs. Filtered simulation shows the concave, convex and sharp Liljequist components from these crystals. The last one makes two arcs, one outside, one inside counterparhelia.

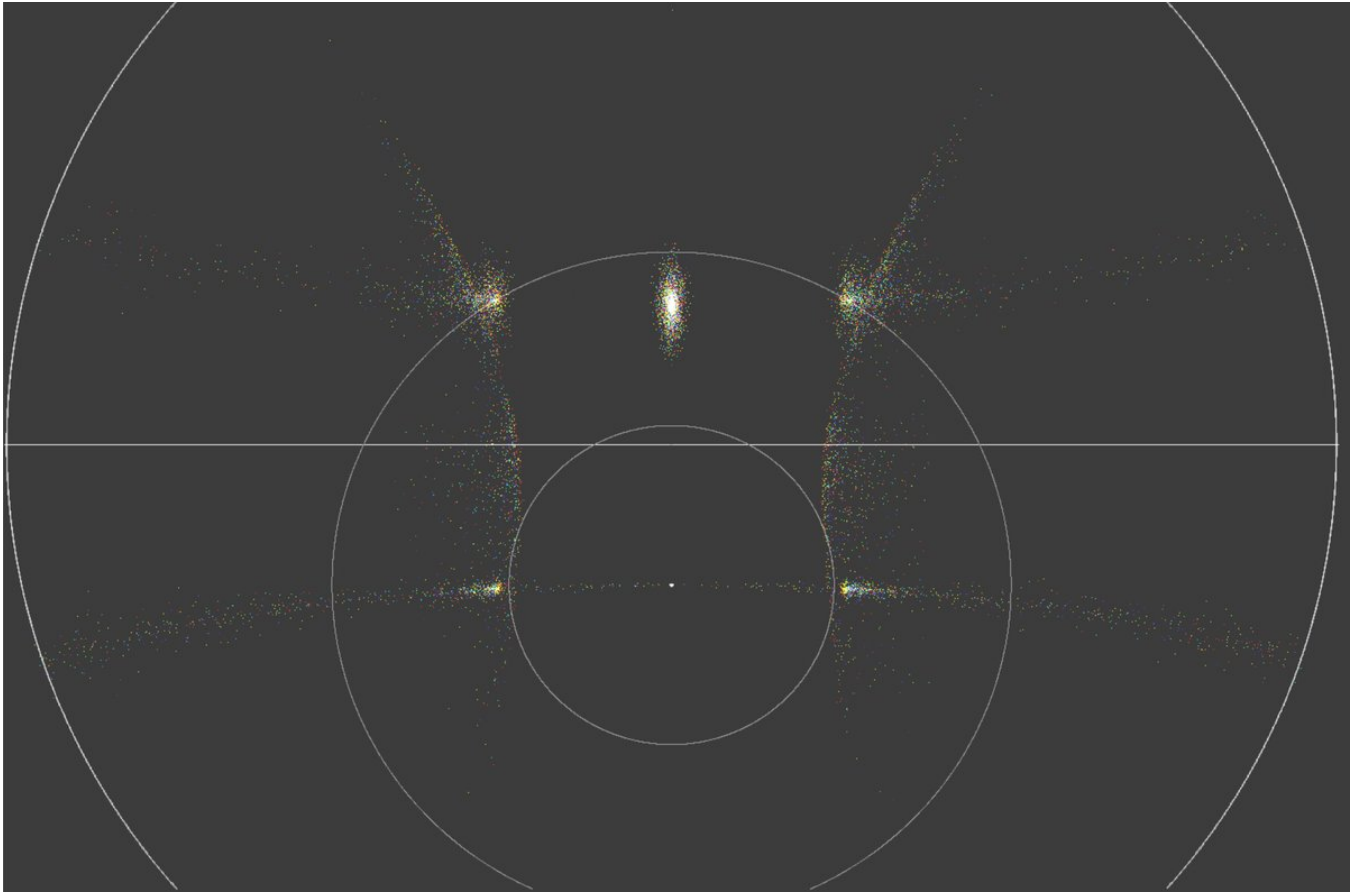
Well, it is not much worth, for one, there should be quite some Bottlinger. There may be something on the sides of the countersun as shown by the last image which is the full 22 frame stack, the other version enhanced by usming. But it is faint and not the right shape. Probably an artefact caused by the

countersun.









21/21 - 2026-01-26 13:24 UTC

9/10 Jan 2026, Rovaniemi. A filtered simulation of "alternate 35° Liljequist arcs". The arcs are less tilted than from normal Lowitz orientation, which may be viewed as an improvement, but the shape is still not right (the reference display: <https://t.co/w4MZsmHixT>). And here, too, the full simulation has unwanted stuff, as shown by the second image – most notably the "alternate 35° Lowitz arcs". But then again many Liljequist arc displays we don't have seem to have Lowitz arcs. So there is that.

